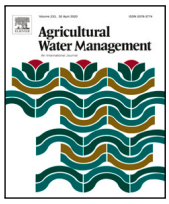
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Economic valuation of irrigation water using production functions: A case study from the Pennaiyar River Basin, Tamil Nadu, India

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ABSTRACT

This study estimates the economic value of irrigation water in the Pennaiyar River Basin, India, using three production function based approaches: the Residual Value Method, Cobb–Douglas, and Translog functions. The analysis is based on farm level data covering five agricultural years from 2017–18 to 2021–22 for paddy, sugarcane, and other crops across seven districts. The dataset includes input cost components and constructed irrigation water volumes, which are used to quantify the marginal contribution of irrigation water and examine its variation across districts and time periods. Higher economic values are observed in Villupuram and Krishnagiri, where coordinated use of water, labour, and machinery increases productivity. Lower values in Cuddalore correspond to high rainfall conditions, where irrigation contributes less to output. The Cobb–Douglas model provides stable elasticity estimates for water valuation, while the Translog specification captures interaction effects between inputs that influence water productivity. The residual method produces higher estimates under low input and high rainfall conditions, reflecting its reliance on residual output after accounting for observed costs. These results indicate that irrigation water value depends on rainfall conditions, input combinations, and production structure. The findings support targeted policy actions, including the implementation of volumetric pricing in districts with higher marginal water values and investment in irrigation infrastructure, canal management, and groundwater recharge in districts with lower values. The estimated values represent marginal economic values derived from production relationships and do not correspond to observed market prices or administratively determined tariffs.

1. Introduction

Irrigation water plays a central role in sustaining agricultural production. It supports food security, contributes to rural livelihoods, and underpins regional development (Namara et al., 2010). Agriculture accounts for around 72% of global freshwater withdrawals, with even higher shares in water-scarce regions such as South Asia (Fao et al., 2018; Berbel et al., 2019; FAO, 2022). Population growth, rapid urban expansion, and climate change continue to place increasing pressure on irrigation systems. These pressures have made water scarcity more widespread and have contributed to more frequent droughts and other extreme events (Wada et al., 2010; Damania, 2020). In India, this challenge is acute. The country supports over 18% of the world's population but has access to only 4% of its renewable freshwater resources (National Water Policy, 2012).

In this context, a clear understanding of the economic value of irrigation water is essential for promoting efficient use and equitable allocation. In many developing countries, including India, irrigation water is supplied through public systems at subsidised or nominal charges, and formal water markets are limited. The National Water Policy (2012) defines water as an economic good and proposes volumetric pricing. However, implementation remains incomplete. Water governance is a state subject, and regulatory enforcement differs across states. Although several policy initiatives and regulatory measures have been introduced to improve groundwater management and water-use efficiency, irrigation pricing continues to follow area-based systems with low cost recovery. This indicates a clear gap between policy provisions and actual practice (Young, 2005; Molle and Berkoff, 2007). As a result, there is often a disconnect between water's actual economic contribution and the price paid by users. This gap discourages efficient use and limits investment in irrigation infrastructure. Recognising this, India's National Water Policy (2012) calls for volumetric pricing, improved cost recovery, and greater farmer participation in irrigation management (National Water Policy, 2012).

Where market prices are unavailable, the economic value of irrigation water must be estimated using indirect methods. These methods

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are classified as stated or revealed preference approaches. Stated preference methods, including contingent valuation and choice experiments, use survey-based responses to estimate the value assigned to irrigation water under hypothetical conditions (Varian, 1992; Hynes et al., 2011; Rolfe and Windle, 2011). Revealed preference methods, such as the travel cost and hedonic pricing approaches, infer value based on observed behaviour in related markets, such as land prices or travel distance to water sources (Young, 1996; Bashe et al., 2022).

Although these methods are used to estimate the value of domestic and recreational water use, their application in agricultural production systems remains limited. Irrigation water is used in combination with other inputs — such as labour, fertilisers, and machinery — to produce crops. This interdependence makes it difficult to isolate water's contribution using preference-based techniques (Young, 2005; Hellegers and Davidson, 2010).

In contrast, production function-based methods are designed to capture the value of water as a productive input (Young, 2005). These methods are classified into deductive and inductive approaches. Deductive methods, such as the Residual Value Method (RVM), assume that farmers maximise profit and assigns the remaining value of output to water after subtracting the cost of other inputs (Hellegers and Davidson, 2010; Berbel et al., 2011). Inductive methods use farm-level data to estimate the marginal contribution of each input, including water. Within this category, the Cobb–Douglas and Translog production functions are among the most common approaches used in agricultural production analysis. These models estimate input elasticities and capture interactions among inputs (Coelli et al., 2005; Buttinelli et al., 2024; Bierkens et al., 2019).

Production function-based approaches are suitable for the Indian context, where irrigation pricing is limited, water is managed through public systems, and agricultural data are available. These approaches estimate the contribution of irrigation water to crop output while accounting for variation in input use and production conditions.

The relationship between irrigation infrastructure and water valuation has direct implications for policy design. Previous studies report that irrigation investment is associated with higher agricultural productivity and rural incomes (Fan et al., 2000; Rosegrant and Evenson, 1992). Empirical evidence also shows that access to irrigation reduces production risk and supports income stability in water-scarce regions (Hussain and Hanjra, 2004; Bhattarai et al., 2002). However, this literature focuses on the expansion of irrigation infrastructure and does not distinguish between alternative system configurations.

Evidence on irrigation system performance indicates that irrigation outcomes depend on the interaction between storage, conveyance, and distribution systems (Molden et al., 2010). Where these components enable controlled water delivery and coordinated input use, irrigation contributes more to output. In contrast, where conveyance is limited or storage is fragmented, the contribution of irrigation water remains constrained.

These differences determine the applicability of policy instruments. Pricing mechanisms, including volumetric allocation, require reliable delivery systems and measurement capacity to influence water use. In regions where irrigation infrastructure is incomplete or water delivery is not controlled, investments in canal networks, tank rehabilitation, and water distribution systems are required to improve irrigation performance (Hussain and Hanjra, 2004; Molden et al., 2010). In contrast, where infrastructure supports reliable and measured water delivery, pricing-based instruments can align water use with its marginal productivity. This indicate that irrigation policy should reflect infrastructure conditions and the economic value of water across regions.

Against this background, this study estimates the economic value of irrigation water in the Pennaiyar River Basin (PRB), India, using farm-level data. The analysis covers five agricultural years and includes three crop groups: paddy, sugarcane, and other major crops. The study applies three production function-based approaches — the Residual Value Method (RVM), Cobb–Douglas, and Translog — to evaluate how model specification affects estimates of water value and the contribution of irrigation within agricultural production.

The study addresses the following objectives: (i) to estimate the economic value of irrigation water across districts and crops using farm-level data; (ii) to compare alternative production function approaches and assess how model specification influences valuation outcomes; and (iii) to examine how variation in rainfall, input use, and cropping patterns affects the estimated value of irrigation water across space and time.

This study contributes to the literature on irrigation water valuation in three ways. First, it provides district-level estimates of the marginal economic value of irrigation water using farm-level data, allowing identification of spatial variation in water productivity within a river basin. Second, it applies three production-based approaches — the Residual Value Method, Cobb–Douglas, and Translog — within a consistent empirical framework, enabling comparison of valuation outcomes under alternative model specifications. Third, it demonstrates that the economic value of irrigation water depends on rainfall conditions and the coordination of complementary inputs, rather than water availability alone. From an international perspective, this study extends existing research that focuses on aggregate or basin-level estimates by providing evidence on intra-regional variation in irrigation water productivity. The findings indicate that uniform pricing and allocation policies may not reflect local production conditions, which has implications for irrigation management in other water-scarce regions with heterogeneous agro-ecological systems.

2. Methods

2.1. Analytical framework

This study applies three production function-based approaches to estimate the economic value of irrigation water: the Residual Value Method (RVM), the Cobb–Douglas production function, and the Translog production function. The RVM derives the value of irrigation water as the residual share of output after accounting for non-water input costs. The Cobb–Douglas and Translog models estimate the marginal value of water using econometric relationships between inputs and output. These approaches differ in functional form and parameter structure, and together provide a basis for comparing estimates of irrigation water value.

2.2. General production function

Agricultural production can be described using a general input–output relationship:

$$Y = f(H, C, M, K, V) \tag{1}$$

where Y denotes the value of agricultural output, and H, C, M, and K represent human labour, animal labour, machinery, and material inputs, respectively. Irrigation water is represented by V, defined as the volume of water applied per unit of cultivated area (m^3/m^2).

All monetary variables are expressed in Indian Rupees per square metre (Rs/m^2). Since the original data are reported in hectares (ha), all area measures are converted to square metres (m^2) using $1\,ha = 10,000\,m^2$. This conversion ensures consistency in unit specification across all input and output variables.

Agricultural output is defined as:

$$Y = \frac{\text{Main Product} + \text{By Product}}{\text{Crop Area}} \tag{2}$$

Human labour is defined as:

$$H = \frac{\text{Family Labour} + \text{Attached Labour} + \text{Casual Labour}}{\text{Crop Area}} \tag{3}$$

Animal labour is defined as:

$$C = \frac{\text{Hired Animal Labour} + \text{Owned Animal Labour}}{\text{Crop Area}} \tag{4}$$

Machinery is defined as:

$$M = \frac{\text{Hired Machine} + \text{Own Machine} + \text{Own Irrigation Machine} + \text{Hired Irrigation Machine}}{\text{Crop Area}} \quad (5)$$

Material inputs are defined as:

$$K = \frac{\text{Seed Value} + \text{Total Fertiliser} + \text{Manure} + \text{Insecticides} + \text{Miscellaneous Cost}}{\text{Crop Area}} \quad (6)$$

These definitions correspond to the cost components recorded in the crop budget data and provide a consistent basis for constructing per unit input measures used in the analysis.

The economic value of irrigation water, denoted as E_w, is estimated using two approaches. The first is the Residual Value Method, which attributes the net value of output to water after deducting the costs of all other inputs. The second is the marginal value product (MVP) approach, where econometric models estimate the additional output generated by an incremental increase in water use, holding other inputs constant.

Each of these approaches builds on Eq. (1) but differs in assumptions and estimation strategy.

2.3. Residual value method (RVM)

The Residual Value Method (RVM) is a deductive approach used to estimate the economic value of irrigation water in the absence of observed market prices. The method is based on the principle that, once the contribution of all other production inputs is accounted for, the remaining value of output can be attributed to irrigation water (Young, 2005).

The economic value of irrigation water is defined as:

$$E_w = \frac{Y - (H + C + M + K)}{V} \quad (7)$$

In Eq. (7), Y denotes the value of agricultural output per unit area, while H, C, M, and K represent the costs of human labour, animal labour, machinery, and material inputs, respectively. The term $Y-(H+C+M+K)$ represents the net value of output after accounting for all non-water input costs. Dividing this residual by V, the volume of irrigation water applied per unit area, yields the economic value of water expressed in Rs/m^3.

The non-water input costs included in $(H + C + M + K)$ are constructed from crop budget data and correspond to the observed production cost components recorded for each district and crop. These include labour costs disaggregated by employment type, machinery costs disaggregated by ownership and usage, and material input costs covering seeds, fertilisers, manure, and plant protection chemicals. All components are expressed on a per unit area basis, ensuring consistency with the production function framework.

This approach assumes that producers maximise profits, that the costs of all non-water inputs are observed, and that input markets operate under competitive conditions (Rodrigues et al., 2021). Under these assumptions, the residual value of output represents the contribution of irrigation water.

This method is based on the Product Exhaustion Theorem, which states that the total value of output is distributed across production inputs according to their marginal contributions (Hellegers and Davidson, 2010). The residual term therefore provides an estimate of the value of the unpriced input.

In this study, the RVM is implemented using observed input–output data and crop-specific water requirements. Although the method provides a transparent estimate of water value, it depends on the completeness and accuracy of input cost data. If relevant cost components are omitted or measured with error, the residual term overstates the contribution of irrigation water. The approach relies on input substitutability, which does not hold across heterogeneous production conditions.

2.4. Cobb–Douglas production function

The Cobb–Douglas production function is a standard approach in agricultural economics because it provides a direct representation of input–output relationships. The model assumes a unitary elasticity of substitution between inputs, where one input can substitute for another at a constant proportion without changing output levels. This assumption describes the substitution relationship among inputs rather than their individual contributions to production.

Each input is assigned a constant output elasticity, denoted by α_i, which represents the percentage change in output resulting from a one percent change in the respective input, holding other inputs constant. Each elasticity parameter α_i represents the proportional contribution of the corresponding input to output. For example, a value of $\alpha_v = 0.2$ indicates that a one percent increase in irrigation water is associated with a 0.2 percent increase in output, holding other inputs constant. This interpretation provides a basis for assessing the contribution of irrigation water within the production process. These elasticities differ across inputs and reflect their contribution to total output. The elasticity of substitution remains constant across input pairs (Cobb and Douglas, 1928).

The Cobb–Douglas production function used in this study is given by:

$$Y = A \cdot H^{\alpha_h} \cdot C^{\alpha_c} \cdot M^{\alpha_m} \cdot K^{\alpha_k} \cdot V^{\alpha_v} \quad (8)$$

where A denotes total factor productivity, which represents the effect of factors not included as explicit inputs in the model. The parameters $\alpha_h, \alpha_c, \alpha_m, \alpha_k, \alpha_v$ represent the output elasticities of human labour, animal labour, machinery, material inputs, and irrigation water, respectively.

To estimate the model parameters, the function is expressed in log-linear form as:

$$\ln Y = \ln A + \alpha_h \ln H + \alpha_c \ln C + \alpha_m \ln M + \alpha_k \ln K + \alpha_v \ln V + \epsilon \quad (9)$$

This transformation enables estimation using the ordinary least squares (OLS) method. The estimation is based on a consistent district–crop level dataset constructed over multiple years. Irrigation input is available for all observations through the constructed volume measure. The model exploits variation across districts and time to identify input elasticities within the Cobb–Douglas framework. The coefficient α_v obtained from the regression represents the elasticity of output with respect to irrigation water. To determine the economic value of water, the marginal value product (MVP) is derived by differentiating the production function with respect to water. Since output is already expressed in monetary terms, this derivative gives the MVP directly:

$$MVP_V = \frac{\partial Y}{\partial V} = \alpha_v \cdot \frac{Y}{V} \quad (10)$$

This expression indicates that the marginal value of irrigation water depends on both its estimated elasticity α_v and the average productivity of water $\frac{Y}{V}$. This expression relates the estimated elasticity to economic valuation by combining the proportional effect of water (α_v) with the observed output-to-water ratio. It represents the value of output associated with an additional unit of irrigation water, given the observed levels of other inputs.

The value of α_v is obtained from the estimated log-linear model. The MVP_V is calculated using observed data on output and water input and is reported in Rs/m^3 for each crop and district. While the Cobb–Douglas model supports straightforward interpretation and efficient estimation, its assumptions impose limitations. The model excludes interaction effects among inputs and restricts substitution patterns to a fixed form. As a result, the model does not capture interaction effects among inputs or variation in substitution patterns within agricultural production systems. The Cobb–Douglas model is used as a baseline, while interaction effects are analysed using the Translog specification, which provides a flexible representation of input relationships.

2.5. Translog production function

The Translog (transcendental logarithmic) production function extends the Cobb–Douglas specification by allowing input elasticities and substitution patterns to vary with input levels. While the Cobb–Douglas function imposes constant elasticities and a fixed elasticity of substitution, the Translog formulation relaxes these restrictions through the inclusion of second-order terms. These terms capture curvature in the production surface and allow interactions between inputs, enabling the model to represent varying input relationships (Veninga, 2017).

The functional form used in this study is (Berndt and Christensen, 1973):

$$\ln Y = \beta_0 + \sum_j \beta_j \ln X_j + \frac{1}{2} \sum_j \sum_k \beta_{jk} \ln X_j \ln X_k + \epsilon \tag{11}$$

where $X_j \in \{H, C, M, K, V\}$ denotes the set of production inputs. The indices j and k refer to elements of this set. The first-order terms $\beta_j \ln X_j$ represent the direct contribution of each input to output. The second-order terms $\beta_{jk} \ln X_j \ln X_k$ represent interaction effects between inputs and squared terms when $j = k$. The parameters β_j and β_{jk} denote first-order and second-order effects, respectively.

The first-order coefficients indicate the immediate response of output to changes in individual inputs. The second-order coefficients indicate how the marginal effect of one input changes with the level of another input, capturing complementarity or substitution. When $j = k$, the squared terms represent diminishing or increasing marginal returns to a given input.

This specification allows for non-constant returns to scale and input interactions. To estimate the economic value of irrigation water, the marginal value product (MVP) is obtained by differentiating the log-output function with respect to $\ln V$, yielding the elasticity of output with respect to water:

$$\frac{\partial \ln Y}{\partial \ln V} = \beta_v + \sum_j \beta_{vj} \ln X_j \tag{12}$$

This expression indicates that the elasticity of output with respect to water depends on the levels of other inputs.

Multiplying the elasticity by the average product of water, $\frac{Y}{V}$, gives the marginal value product of water:

$$MVP_V = \left(\beta_v + \sum_j \beta_{vj} \ln X_j \right) \cdot \frac{Y}{V} \tag{13}$$

This formulation combines the direct effect of water with its interaction effects, implying that the economic value of water varies across observations with different input combinations.

The Translog specification increases the number of parameters to be estimated due to the inclusion of interaction terms. In datasets with limited variation or strong correlations among inputs, this results in estimation challenges, including multicollinearity. The interpretation of parameter estimates must account for variation in coefficient signs and statistical significance across model specifications.

2.6. Kolmogorov–Smirnov test for distributional comparison

To evaluate spatial variation in the economic value of irrigation water (E_w) across districts, the Kolmogorov–Smirnov (KS) test is employed. The test compares the empirical cumulative distribution functions (ECDFs) of district-level E_w estimates obtained from the Residual Value Method (RVM), Cobb–Douglas, and Translog specifications. This procedure allows assessment of differences across the full distribution of values rather than using summary statistics alone.

For each pair of districts, the null hypothesis is defined as:

$$H_0 : F_i(x) = F_j(x) \tag{14}$$

$$H_1 : F_i(x) \neq F_j(x) \tag{15}$$

where $F_i(x)$ and $F_j(x)$ denote the ECDFs of irrigation water values for districts i and j, respectively.

The ECDF for a sample of size n is given by:

$$F_n(x) = \frac{1}{n} \sum_{k=1}^{n} \mathbb{I}(X_k \leq x) \tag{16}$$

where X_k represents observed values of E_w. The ECDF describes the cumulative proportion of observations that do not exceed a given value and therefore characterises the distribution of irrigation water values within each district.

The KS statistic is defined as:

$$D = \sup_x \left| F_i(x) - F_j(x) \right| \tag{17}$$

which represents the maximum deviation between the two ECDFs across all values of x. This statistic provides a measure of the extent to which the two distributions differ.

Pairwise KS tests are conducted for all district combinations within each crop, year, and model specification. The resulting p-values indicate whether the distributions of irrigation water values differ across districts. These outcomes are summarised using heatmaps, where each cell represents the result of a pairwise comparison.

The use of the KS framework follows from the observed heterogeneity in E_w distributions across districts. Differences in dispersion, skewness, and range indicate that mean-based comparisons do not capture distributional variation. The KS test, through its reliance on ECDFs, captures shifts in the entire distribution and provides a basis for assessing spatial variation in irrigation water productivity.

2.7. Comparative evaluation of modelling approaches

The three models applied in this study differ in their assumptions, estimation strategies, and data requirements. These differences affect how each model captures irrigation water's economic contribution.

The Residual Value Method (RVM) attributes the remaining output value to water after deducting the cost of all other inputs. It assumes perfect substitutability among inputs and tends to produce higher estimates for water value, as it assigns the full residual share to the unpriced input (Gibbons, 1986; Young, 2005). The method provides a transparent estimation structure, but the resulting values depend on the accuracy of input cost data. In contexts where input subsidies or reporting gaps are common, RVM can overstate water's contribution (Ward and Michelsen, 2000).

The Cobb–Douglas production function offers a parametric structure based on constant output elasticities and a fixed substitution elasticity of one. Its log-linear form allows for straightforward estimation using standard regression techniques. The model produces stable and interpretable results, making it suitable for baseline analysis. However, it does not accommodate interaction effects among inputs or changes in substitution patterns, which can limit its ability to reflect production complexity in diverse agricultural settings (Griliches, 1963).

The Translog model expands on the Cobb–Douglas form by including squared and interaction terms. This structure allows for variable substitution elasticities and captures complementarity or substitutability among inputs. As a second-order approximation to a general production function, it accommodates more realistic representations of input relationships (Christensen et al., 1973; Coelli et al., 2005). However, the model introduces a larger number of parameters, increasing the risk of multicollinearity and requiring a more extensive dataset for reliable estimation (Lovell et al., 1993) (see Table 1).

3. Study area and data requirements

This study focuses on the Pennaiyar River Basin (PRB), located in Tamil Nadu, India. The basin spans seven districts: Cuddalore, Dharma-

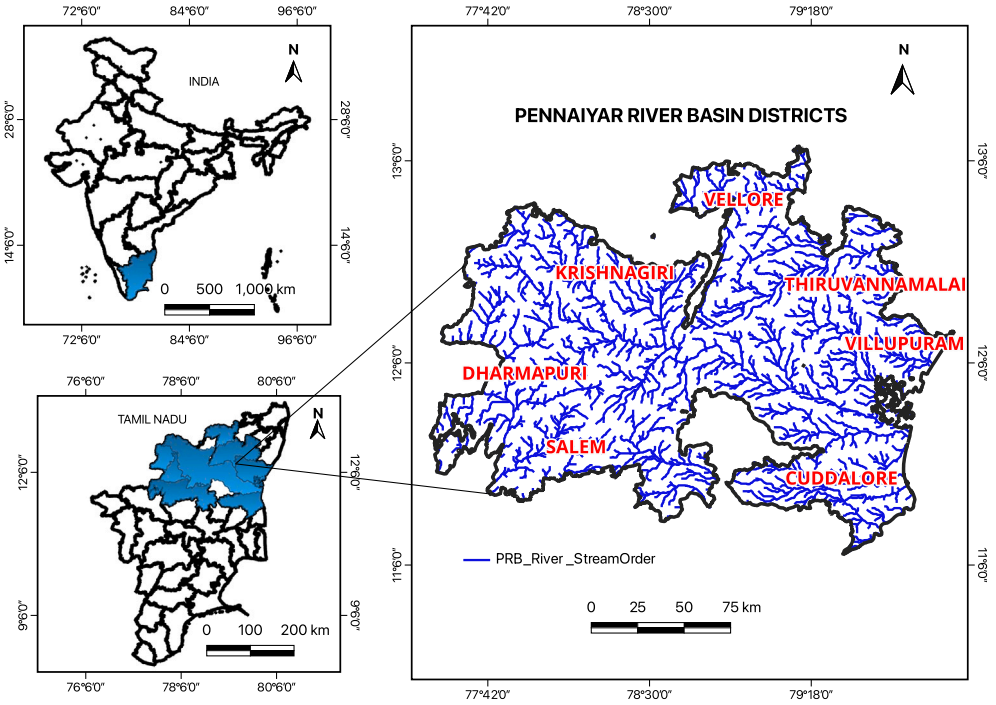


Fig. 1. Study area map of the Pennaiyar River Basin (PRB) in Tamil Nadu, India, showing district boundaries, river networks, reservoirs, and irrigation-related administrative regions.

Table 1
Comparison of the Residual Value Method (RVM), Cobb–Douglas (CD), and Translog approaches for estimating the economic value of irrigation water.

Aspect	RVM	Cobb–Douglas	Translog
Model type	Residual-based	Parametric function	Flexible functional form
Substitution	Perfect	Constant elasticity (unitary)	Variable elasticity
Input interactions	Not captured	Not captured	Captured
Estimation stability	Sensitive to input errors	High	Dependent on data quality
MVP output	Higher (upper bound)	Moderate	Variable; includes negative estimates
Interpretability	Straightforward	Clear and stable	Requires care in interpretation
Limitations	Overstates water value if cost data are inaccurate	Omits interaction and nonlinear effects	Affected by multicollinearity; requires more data

Note. Summary based on assumptions and empirical properties from Young (2005), Heady and Dillon (1961), Christensen et al. (1973), and Coelli et al. (2005).

puri, Krishnagiri, Salem, Thiruvannamalai, Vellore, and Villupuram. Geographically, it extends between latitudes 11°38′30″N and 12°54′00″N, and longitudes 77°39′30″E and 79°54′15″E (Fig. 1). While most of the basin lies within Tamil Nadu, small sections reach into the neighbouring states of Karnataka and Andhra Pradesh. The Tamil Nadu portion of the PRB covers approximately 11,375.55 km^2.

The basin receives an average annual rainfall of 977 mm, with 76% of this total concentrated in the Southwest (June to September) and Northeast (October to December) monsoon seasons (National Water Development Agency, Government of India, 2010). The region's renewable water resources include an estimated 1319.58 million cubic metres (mcm) of surface water and 766.77 mcm of groundwater (National Water Mission, Government of India, 2017).

Agricultural production in the PRB is dominated by paddy and sugarcane, which together account for 3213.94 km^2 or 68.71% of the total cultivated area (4677.55 km^2). Other field crops, including groundnut and cholam, occupy 1264.28 km^2 (27.02%), while plantation crops such as cashew and jackfruit cover 199.33 km^2 (4.26%).

Table 2 summarises the data components and their sources, which span five agricultural years (2017–18 to 2021–22) and include all districts and crop categories.[1] The analysis is based on consistent farm budget records, crop-specific irrigation data, and inflation-adjusted prices, allowing for reliable comparisons across time and location. Irrigation water input, denoted by V, is defined as the volume of water applied per unit of cultivated area (m^3/m^2). The volume is calculated using Volume = Area × Depth, where crop area is converted from hectares to square metres and water depth from millimetres to metres. Irrigation depth is derived from crop water requirements and adjusted using effective rainfall estimates based on district-level rainfall data. This procedure follows standard FAO-based water accounting approaches and ensures that irrigation input is consistently constructed for all observations across districts and time periods. The distribution of observations across districts, crop categories, and years is reported in Appendix (Tables A.13–A.15).

All monetary variables used in the analysis are adjusted for inflation to ensure comparability across years. Nominal values are converted into real terms using the Consumer Price Index (CPI), with 2017–18 specified as the base year. The deflation is implemented as follows:

$$X_t^{real} = \frac{X_t^{nominal}}{\mathrm{CPI}_t} \times \mathrm{CPI}_{base} \tag{18}$$

where $X_t^{nominal}$ denotes the observed value in year t, CPI_t represents the Consumer Price Index for year t, and CPI_{base} corresponds to the index value for the base year (2017–18). This adjustment is applied to all monetary variables, including output and input cost components. As a result, all rupee-denominated variables are expressed in constant prices of the base year, ensuring that variation across years reflects changes in production conditions rather than price effects.

[1] For sugarcane, data were available only up to 2020–21 due to incomplete reporting in 2021–22.

Table 2
Data sources used in the study.

Data	Components	Source
Crop budgets (Desagri 2023, 2023)	Area, output (main and by-products), labour (family, attached, casual), animal and machinery use (owned, hired), inputs (seeds, fertilisers, manure, pesticides, etc.)	Ministry of Agriculture, GoI[1]
Crop water requirements (FAO, 1986; TNAU, 2023)	Irrigation water depth (mm)	FAO[2], TNAU[3]
Rainfall data (TNWRD, 2022)	District-level daily and monthly rainfall	State Ground and Surface Water Resources Data Centre, WRD[4], GoTN[5]
Inflation rates (NSSO, 2023)	Consumer Price Index (CPI; base year 2017–18)	Ministry of Statistics, GoI[1]

[1] GoI – Government of India.
[2] FAO – Food and Agriculture Organisation.
[3] TNAU – Tamil Nadu Agricultural University.
[4] WRD - Water Resources Department.
[5] GoTN – Government of Tamil Nadu.

4. Results

This section presents the main findings in three parts: (i) estimates of input elasticities, (ii) economic valuation of irrigation water, and (iii) spatial and temporal patterns across districts and years. These results highlight substantial variation in the marginal value of irrigation water across crops, models, locations, and time. The observed differences reflect crop-specific input use patterns and production conditions. These findings can provide a sound empirical basis for designing irrigation pricing strategies that respond to local agricultural and economic conditions.

4.1. Model diagnostics

Diagnostic tests were conducted for each crop–year model to evaluate multicollinearity, heteroscedasticity, autocorrelation, and model specification for both the Cobb–Douglas and Translog production functions. The corresponding results are reported in Tables A.6–A.8 and Tables A.9–A.11.

For the Cobb–Douglas specification, multicollinearity was assessed using Variance Inflation Factors (VIF). The estimated VIF values range from 1.097 to 1.362 for paddy, 1.243 to 1.485 for sugarcane, and 1.326 to 2.792 for other crops. These values remain below the threshold of 10 (Kutner et al., 1984) and the threshold of 5 proposed in applied econometric studies (O'brien, 2007). This indicates that multicollinearity does not affect the stability of coefficient estimates. For the Translog specification, VIF is not reported. The inclusion of squared and interaction terms generates deterministic relationships among regressors, which inflate VIF values by construction and limit their interpretability (Greene, 2003; Wooldridge, 2010).

Heteroscedasticity was examined using the Breusch–Pagan (BP) test (Breusch and Pagan, 1979). The null hypothesis of constant variance is rejected for paddy (2018–19 to 2021–22), sugarcane (2017–18 and 2018–19), and other crops (2017–18 and 2021–22). Similar outcomes are observed for the Translog specification in selected years. To address this issue, all models were estimated using heteroscedasticity-consistent (HC1) standard errors, ensuring valid statistical inference.

Model specification was evaluated using the Ramsey Regression Equation Specification Error Test (RESET), which assesses potential functional form misspecification and omitted variable bias (Ramsey, 1969). The null hypothesis of correct functional form is not rejected for most crop–year models. The RESET test p-values range from 0.049 to 0.522 for paddy, 0.019 to 0.602 for sugarcane, and 0.002 to 0.505 for other crops (Tables A.6–A.8). Rejections are observed for paddy (2020–21) and other crops (2018–19, 2019–20, and 2021–22), but these cases are not concentrated within a single crop or time period. These results indicate that the Cobb–Douglas specification provides an adequate representation of the production relationship, and no consistent evidence of functional form misspecification is observed.

Autocorrelation was assessed using the Durbin–Watson test. The results indicate rejection of the null hypothesis of no serial correlation in several crop–year models. The Durbin–Watson test p-values range from 0.000 to 0.005 for paddy, 0.000 to 0.509 for sugarcane, and 0.000 across all years for other crops (Tables A.6–A.8). Evidence of autocorrelation is observed in most paddy models and all other crop models, while sugarcane exhibits mixed results across years. Given that the data consist of cross-sectional observations at the tehsil level for each year, these patterns arise from spatial dependence and unobserved heterogeneity rather than temporal dependence. These results are consistent with cross-sectional dependence across observational units rather than temporal autocorrelation.

Unit root tests were not conducted, as the dataset does not represent a time-series or panel time-series structure. Each year is analysed as an independent cross-section, and stationarity conditions are therefore not required.

Model fit was evaluated using the coefficient of determination (R^2) and adjusted R^2. For the Cobb–Douglas specification, R^2 ranges from 0.181 to 0.456 for paddy, 0.275 to 0.486 for sugarcane, and 0.389 to 0.833 for other crops. The corresponding adjusted R^2 values remain close to the R^2 values, indicating that the models do not suffer from over-parameterisation. The moderate R^2 values obtained for the Cobb–Douglas specification reflect the cross-sectional structure of the dataset and the exclusion of factors such as soil characteristics, rainfall variation, and farm management practices from the estimated model. In this context, the Cobb–Douglas model is used to estimate input elasticities and marginal effects rather than to maximise explanatory power. The estimated coefficients therefore remain informative for evaluating the contribution of irrigation water despite moderate model fit.

The number of observations differs across crop–year combinations, with smaller sample sizes reported for sugarcane in selected years. Given the larger number of parameters in the Translog specification, the reduced sample size lowers the degrees of freedom and affects statistical significance. The results are therefore interpreted based on consistent patterns across crops and years rather than isolated coefficient estimates.

The diagnostic results support the adequacy of the estimated models. Multicollinearity is not evident, heteroscedasticity is addressed through robust estimation, and specification tests indicate an appropriate functional form in most cases. Although autocorrelation is detected in selected models, it does not affect the interpretation of the estimated elasticities. These results support the use of the estimated coefficients for deriving the marginal value product of irrigation water.

4.2. Model-based estimation of input contributions

First we study the relative contribution of key production inputs—human labour, animal labour, machinery, materials, and irrigation water—using estimated coefficients from the Cobb–Douglas and Translog production functions. The analysis spans three crop categories over five agricultural years. Table 3 provides an overview of the regression specifications applied to each crop, including the models used, time periods covered, and links to the corresponding coefficient estimates. Complete regression outputs are available in Appendix A.1.

Across the three functional forms, human labour records the largest and most consistent contribution to output. Positive and significant coefficients ($p< 0.01$) are observed in more than 85% of crop–year combinations. This pattern reflects the dependence of crop production

Table 3
Crop-wise regression estimates under the Cobb–Douglas and Translog production functions across agricultural years.

Crop	Model	Year	Coefficient table
Paddy	Cobb–Douglas	2017–18 to 2021–22	Table A.6
Sugarcane	Cobb–Douglas	2017–18 to 2020–21	Table A.7
Other crops	Cobb–Douglas	2017–18 to 2021–22	Table A.8
Paddy	Translog	2017–18 to 2021–22	Table A.9
Sugarcane	Translog	2017–18 to 2020–21	Table A.10
Other crops	Translog	2017–18 to 2021–22	Table A.11

Table 4
Annual rainfall across districts from 2017–18 to 2021–22.

District	2017–18	2018–19	2019–20	2020–21	2021–22
Cuddalore	1032	1145	998	810	1179
Dharmapuri	1004	987	1012	789	1044
Krishnagiri	1076	1110	1030	856	1121
Salem	1018	1029	987	808	1099
Thiruvannamalai	1056	1074	1010	823	1132
Vellore	1098	1102	1045	841	1150
Villupuram	1024	1088	999	832	1116
Basin Avg. (mm)	**1044**	**1078**	**1011**	**821**	**1120**

on manual labour, with paddy cultivation requiring substantial labour input for transplanting and harvesting operations.

A deviation from expected patterns occurs for machinery in paddy cultivation during 2017–18, where the estimated coefficient is negative and significant. This result indicates that mechanisation did not contribute to output under the observed production conditions.

In smallholder agricultural systems, production is carried out on fragmented landholdings where scale economies are limited. Under such conditions, machinery use can increase production costs without proportional gains in output. This arises from underutilisation of equipment, mismatch between machine capacity and farm size, and reliance on hired machinery services that are not aligned with crop operations (Binswanger, 1978; Pingali, 2007).

The negative coefficient also captures the influence of omitted factors correlated with machinery use. Substitution towards mechanisation among farms with labour constraints or lower soil quality biases the estimated coefficient downward. Evidence from developing-country agriculture shows that mechanisation does not generate productivity gains in the absence of complementary inputs and effective management practices (Diao et al., 2014, 2016).

This result is consistent with the observed production structure in the study area, where mechanisation is not complemented by proportional adjustments in other inputs.

Irrigation water also showed a measurable effect in many models for paddy and other crops ($p< 0.05$). However, the elasticity estimates obtained from the Cobb–Douglas model range from 0.05 to 0.15 in most crop–year combinations. These values indicate that a 1% increase in water use is associated with a 0.05% to 0.15% increase in output, indicating a low marginal contribution of irrigation water. This pattern reflects limitations in irrigation application and the mismatch between water use and crop water requirements (Seckler et al., 1998; Kijne et al., 2003). In sugarcane models, water elasticity values remain lower and are insignificant in several crop–year combinations. This result is consistent with the longer growth period of sugarcane and its capacity to tolerate short-term moisture deficits (Khuimphukhieo et al., 2024).

The Translog model captured a wider range of production behaviour, including nonlinear effects and interactions among inputs. In several instances, the squared term for irrigation water was negative and statistically significant. For example, in paddy (2020–21), the estimate for the squared water term was $\beta_{\text{Water}^2} = -0.038$ ($p< 0.05$), indicating diminishing returns to water input at higher usage levels.

Interaction terms provided further insight into input coordination. In paddy production during 2019–20, the interaction between water and machinery was positive and significant ($\beta = 0.022$,$p< 0.05$), suggesting that access to mechanisation improved irrigation efficiency. Positive water–labour interactions were also observed in multiple years, indicating that input coordination contributed to improved outcomes. In contrast, interaction terms in sugarcane models were less prominent, implying limited complementarity between water and other inputs.

Although irrigation water remains an important input, its effectiveness depends on how it is used alongside other resources. Enhancing irrigation productivity therefore requires integrated input planning, including access to labour, mechanisation, and appropriate crop management practices tailored to local conditions.

4.3. Temporal variation in water input coefficients

Assessing how the contribution of irrigation water changes over time is important for understanding its role under varying rainfall and production conditions. The estimated coefficients from the Cobb–Douglas and Translog models highlight differences in how irrigation water affects crop output across years (see Fig. 2).

The Cobb–Douglas model produced a narrow range of estimated water input elasticities, between 0.05 and 0.25. This consistency simplifies interpretation and enables straightforward comparison across years and crops. However, the model's fixed elasticity assumption constrains its capacity to reflect year-to-year changes in water use efficiency. Since the relationship between inputs and output is assumed to remain constant, the model cannot account for temporal variation in irrigation effectiveness arising from differences in rainfall patterns, crop stages, and management practices. For sugarcane, the model returned near-zero water elasticity values in all years, despite documented variation in irrigation demand due to its extended growth duration and season-dependent water requirements (FAO, 2023).

In contrast, the Translog model revealed wider year-to-year variation in irrigation responsiveness. For paddy, the coefficient declined to −0.2 in 2019–20 and increased to 2.2 in 2020–21. This change aligns with a reduction in mean basin rainfall from 1028 mm in 2019–20 to 821 mm in 2020–21, a drop of 20% compared to the five-year average (2017–2021: 1027 mm). Lower rainfall in 2020–21 increased irrigation requirements, resulting in a greater marginal contribution of water to output.

A similar pattern was observed for sugarcane. The water coefficient reached 1.8 in 2018–19 and dropped to −2.1 in 2020–21. These shifts correspond to years when rainfall fell below the basin average, which have affected irrigation schedules and reduced water-use efficiency. For other crops, the coefficient rose from −2.3 in 2017–18 to 1.6 in 2021–22. This trend corresponds with changes in cropping patterns and improved coordination between water and other inputs such as labour and machinery.

These trends are supported by the rainfall statistics shown in Table 4, which identify 2020–21 as the driest year across all districts. In that year, the Translog model consistently produced the highest irrigation water coefficients. This association suggests that water becomes more economically productive when rainfall is insufficient to meet crop needs.

The results underscore the limitations of models with fixed elasticity assumptions, such as Cobb–Douglas, in capturing inter-annual variability. The Translog model, by incorporating interaction effects and nonlinearities, offers a more detailed account of irrigation's changing role across years.

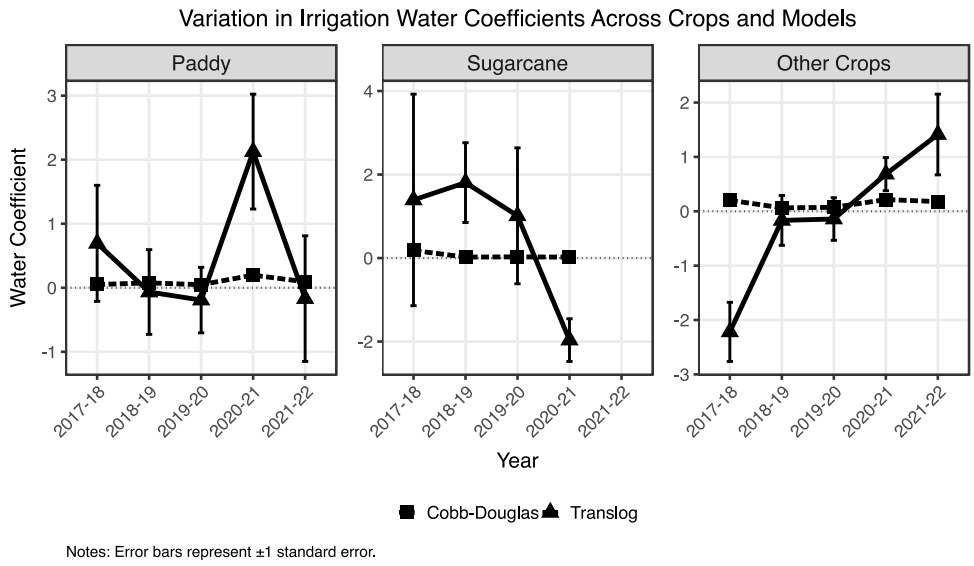


Fig. 2. Irrigation water coefficients for paddy, sugarcane, and other crops under the Cobb–Douglas (CD) and Translog production functions across agricultural years. Error bars represent ±1 standard error.

4.4. Economic value of irrigation water

This section presents district-level estimates of the economic value of irrigation water (Rs/acre).[2] Fig. 3 illustrates temporal trends for each crop under the three production models (RVM, Cobb–Douglas, and Translog). Table 5 summarises mean values and standard deviations by district and model.

Comparisons across districts indicated substantial differences in the economic value of irrigation water among crops. Sugarcane consistently produced higher returns compared to paddy and other crops. For instance, the Translog estimates for sugarcane peaked at Rs. 24.49/acre in 2017–18, exceeding the highest values obtained for paddy (Rs. 4.12/acre) and other crops (Rs. 3.42/acre) in the same model. This higher economic value reflects sugarcane's greater water requirements and longer cultivation cycle, enabling the crop to generate increased returns from irrigation.

In contrast, paddy generated lower economic returns from irrigation water compared to sugarcane across all districts and models. For example, the highest RVM value recorded for paddy was Rs. 1.82/acre (2017–18 and 2019–20), which remained below sugarcane's corresponding values of Rs. 8.28/acre and Rs. 9.62/acre. Likewise, under the Cobb–Douglas model, paddy's highest estimate (Rs. 0.90/acre in 2020–21) was lower than the maximum estimate for sugarcane (Rs. 3.13/acre in 2017–18). These results suggest that paddy cultivation provides limited economic returns per unit of irrigation water, especially in water-scarce districts such as Krishnagiri and Dharmapuri.

Other crops, including groundnut, pulses, and cotton, provided moderate economic returns that improved over time. For instance, under RVM, values increased from Rs. 1.39/acre (2017–18) to Rs. 2.56/acre (2021–22). Under the Translog model, other crops shifted from negative values (e.g., –Rs. 2.96/acre, 2017–18) to positive returns (Rs. 3.42/acre, 2020–21). These improvements reflect better crop selection and more effective input management, in districts like Villupuram and Thiruvannamalai, where input complementarity was higher.

Comparisons across districts indicate that sugarcane cultivation generates higher economic returns per unit of irrigation water compared to paddy and other crops. Despite these higher returns, sugarcane requires substantial irrigation, limiting its suitability in districts facing water scarcity. Negative estimates of the marginal value of irrigation water under the Translog specification (e.g., sugarcane in 2019–20 and 2020–21) indicate that, at observed input levels, additional water reduces the value of output. This result reflects the structure of the Translog function, in which interaction and squared terms allow the marginal contribution of water to vary with the levels of other inputs. When water application exceeds the level supported by complementary inputs such as labour and material inputs, the estimated marginal value becomes negative, indicating diminishing returns to irrigation. From a policy perspective, these estimates identify conditions in which irrigation use exceeds economically efficient levels. This provides evidence for targeted water management interventions, including the adoption of water-saving practices, improved scheduling of irrigation, and better coordination between water use and input application. In contrast, the consistent improvement in economic water values observed for other crops supports crop diversification as a viable alternative. Such diversification can enhance economic sustainability in districts experiencing frequent water constraints.

4.5. District-level differences in irrigation water economic value

Irrigation water values vary across districts in the Pennaiyar River Basin. These differences are associated with rainfall patterns, irrigation structures, crop composition, and the coordination of production inputs. Figs. 4, 6, and 7 present Kolmogorov–Smirnov (KS) heatmaps that summarise the statistical significance of distributional differences between districts for paddy, sugarcane, and other crops, based on the Residual Value Method (RVM), Cobb–Douglas, and Translog models. The shading reflects the strength of significance: ns ($p \geq 0.05$), * ($0.01 \leq p < 0.05$), ** ($0.001 \leq p < 0.01$), and *** ($p < 0.001$). Full pairwise results are reported in Appendix A.2.

To illustrate these distributional differences, Fig. 5 presents selected ECDF comparisons involving Cuddalore, which shows strong contrasts with Villupuram, Thiruvannamalai, and Dharmapuri.

In Fig. 5(a), the KS statistic ($D = 0.243$) indicates a clear distributional difference between Villupuram and Cuddalore for other crops in 2019–20. A comparable difference is observed in Fig. 5(b), where paddy values in Thiruvannamalai differ from those in Cuddalore ($D = 0.234$). The largest deviation is observed in Fig. 5(c), where sugarcane outcomes in Dharmapuri diverge from those in Cuddalore ($D = 0.375$).

[2] Economic water values (Rs/acre) were calculated as $E_w^{\text{acre}} = E_w^{\text{m}^3} \times \text{Volume}_{\text{acre}}$, where $E_w^{\text{m}^3}$ is the estimated economic value per cubic metre and $\text{Volume}_{\text{acre}}$ represents the district- and crop-specific irrigation volume per acre (m^3).

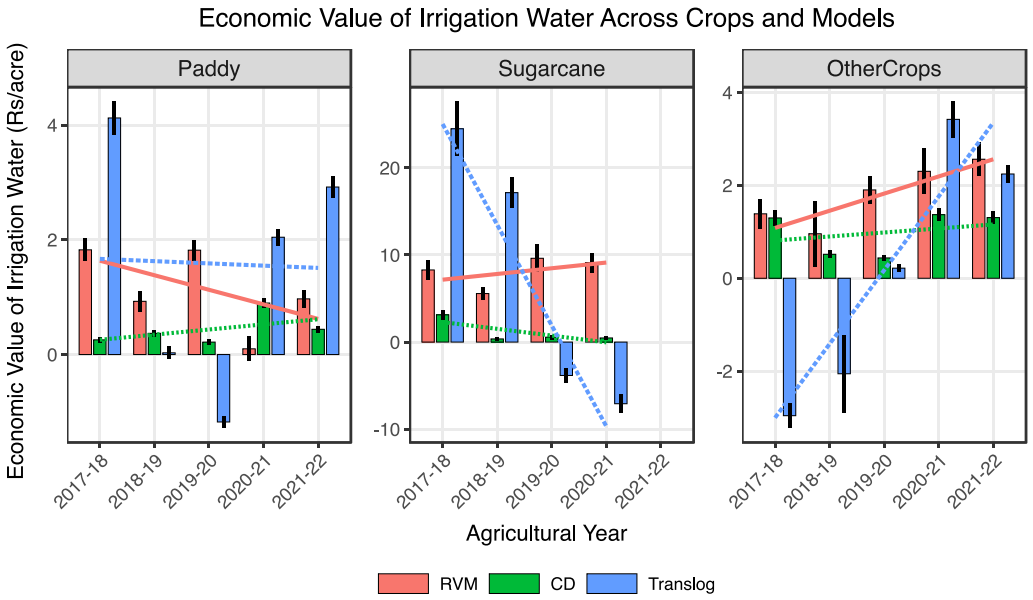


Fig. 3. Economic value of irrigation water (Rs/acre) for paddy, sugarcane, and other crops under the Residual Value Method (RVM), Cobb–Douglas (CD), and Translog production functions across agricultural years. Bars represent mean economic values, and error bars represent ±1 standard error. Trend lines indicate temporal variation in estimated irrigation water values for each production model.

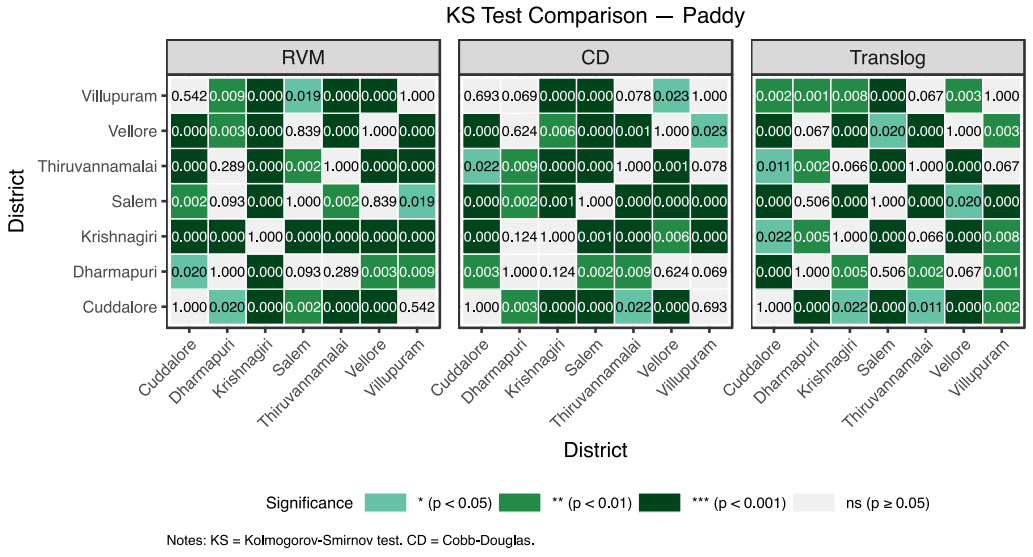
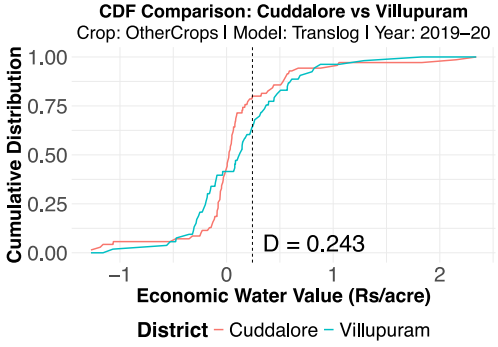


Fig. 4. Kolmogorov–Smirnov (KS) test heatmap for paddy, showing pairwise statistical comparisons of economic irrigation water value distributions across districts under the Residual Value Method (RVM), Cobb–Douglas (CD), and Translog production functions. Cell values represent KS test p-values, and colour intensity indicates the level of statistical significance between district-level distributions.

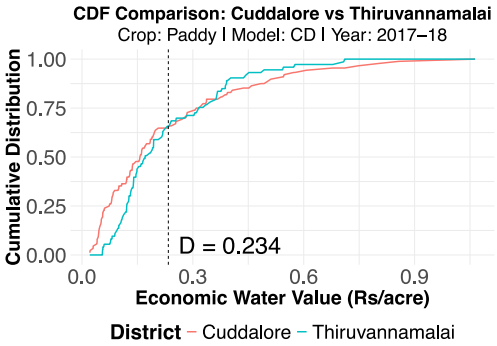
Villupuram records the highest irrigation water values across crops and models. Under the Translog model for paddy (Fig. 4), the estimated value in 2017–18 is Rs. 4.12/acre (SD = 3.70), with rainfall of 1024 mm. This outcome reflects alignment between rainfall timing, irrigation availability, and input deployment. Statistically significant differences are observed between Villupuram and Cuddalore, Krishnagiri, and Vellore (***), indicating distinct production structures.

In contrast, Dharmapuri and Salem exhibit lower and more concentrated distributions of irrigation water values. For sugarcane in Dharmapuri during 2019–20, the Cobb–Douglas estimate is Rs. 0.56/acre (SD = 0.44), with rainfall of 1012 mm (Fig. 6). These values indicate limited variation in input use and weaker responsiveness to rainfall conditions. Fewer statistically significant differences are observed, except in comparisons with Villupuram and Thiruvannamalai.

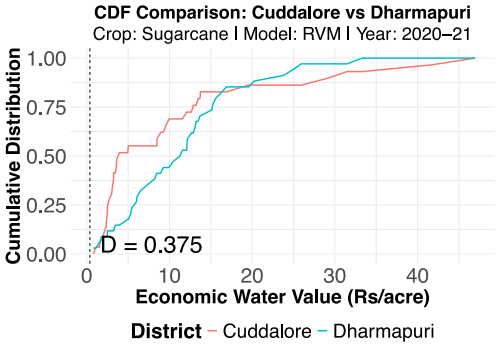
In Krishnagiri, paddy irrigation values differ from those in Cuddalore under the RVM model (2017–18), despite similar mean values. The KS statistic ($D = 0.73$) indicates a substantial distributional difference. This reflects more controlled irrigation conditions in Krishnagiri (942 mm rainfall) relative to Cuddalore, where higher rainfall reduces dependence on irrigation.



(a) Other crops under the Translog production function (2019–20): Cuddalore vs Villupuram



(b) Paddy under the Cobb–Douglas (CD) production function (2017–18): Cuddalore vs Thiruvannamalai



(c) Sugarcane under the Residual Value Method (RVM) (2020–21): Cuddalore vs Dharmapuri

Fig. 5. Empirical cumulative distribution function (ECDF) comparisons of irrigation water values between districts for different crops, production models, and agricultural years. The dashed vertical line in each panel represents the Kolmogorov–Smirnov (KS) test statistic, which measures the maximum difference between cumulative distributions.

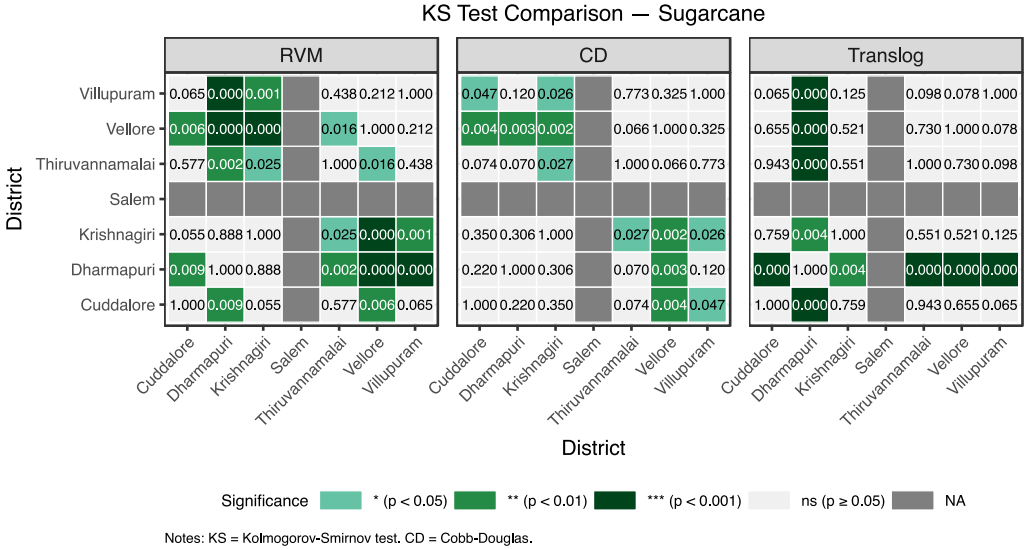


Fig. 6. Kolmogorov–Smirnov (KS) test heatmap for sugarcane, showing pairwise statistical comparisons of economic irrigation water value distributions across districts under the Residual Value Method (RVM), Cobb–Douglas (CD), and Translog production functions. Cell values represent KS test p-values, and colour intensity indicates the level of statistical significance between district-level distributions.

Table 5
Economic value of irrigation water (Rs/acre) by crop, production model, and agricultural year.

Crop	Model	Year	Mean E_w	SD E_w
Paddy	RVM	2017–18	1.825	2.382
Paddy	RVM	2018–19	0.928	1.950
Paddy	RVM	2019–20	1.819	2.330
Paddy	RVM	2020–21	0.100	2.633
Paddy	RVM	2021–22	0.970	1.865
Paddy	CD	2017–18	0.256	0.215
Paddy	CD	2018–19	0.370	0.413
Paddy	CD	2019–20	0.217	0.188
Paddy	CD	2020–21	0.898	0.755
Paddy	CD	2021–22	0.440	0.318
Paddy	Translog	2017–18	4.124	3.699
Paddy	Translog	2018–19	0.030	1.060
Paddy	Translog	2019–20	−1.175	1.176
Paddy	Translog	2020–21	2.045	1.704
Paddy	Translog	2021–22	2.920	2.369
Sugarcane	RVM	2017–18	8.281	8.023
Sugarcane	RVM	2018–19	5.570	4.432
Sugarcane	RVM	2019–20	9.622	10.310
Sugarcane	RVM	2020–21	9.117	8.875
Sugarcane	CD	2017–18	3.128	3.002
Sugarcane	CD	2018–19	0.364	0.292
Sugarcane	CD	2019–20	0.563	0.441
Sugarcane	CD	2020–21	0.475	0.469
Sugarcane	Translog	2017–18	24.490	25.229
Sugarcane	Translog	2018–19	17.152	14.461
Sugarcane	Translog	2019–20	−3.833	4.706
Sugarcane	Translog	2020–21	−7.077	8.045
OtherCrops	RVM	2017–18	1.388	3.907
OtherCrops	RVM	2018–19	0.959	9.907
OtherCrops	RVM	2019–20	1.903	4.016
OtherCrops	RVM	2020–21	2.304	6.336
OtherCrops	RVM	2021–22	2.564	4.354
OtherCrops	CD	2017–18	1.300	1.445
OtherCrops	CD	2018–19	0.517	0.582
OtherCrops	CD	2019–20	0.438	0.442
OtherCrops	CD	2020–21	1.369	1.313
OtherCrops	CD	2021–22	1.306	1.149
OtherCrops	Translog	2017–18	−2.960	3.148
OtherCrops	Translog	2018–19	−2.057	11.795
OtherCrops	Translog	2019–20	0.218	0.712
OtherCrops	Translog	2020–21	3.423	4.905
OtherCrops	Translog	2021–22	2.246	1.924

Cuddalore consistently records lower irrigation water values across crops and models. Higher rainfall reduces the marginal contribution of irrigation, particularly for paddy. Significant KS differences (*** and **) in comparisons with Villupuram and Krishnagiri indicate that irrigation returns depend on differential access to natural water sources.

Thiruvannamalai shows intermediate irrigation values for other crops, particularly under the Translog model (Fig. 7). Statistically significant differences with Cuddalore and Dharmapuri indicate variation in input combinations associated with rainfall conditions and cropping patterns.

Vellore exhibits moderate and stable irrigation water values across crops. The limited number of statistically significant differences indicates a relatively consistent production structure, supported by stable rainfall and input use.

5. Discussion

This study evaluates whether irrigation water can be assigned a uniform economic value across districts. The results reject this proposition. The economic contribution of irrigation water varies across districts because production depends on the structure and performance of irrigation systems. Water valuation is therefore conditional on infrastructure configuration rather than water availability alone.

The variation in economic water values reflects differences in how irrigation systems convert stored water into field-level application. In districts where storage, conveyance, and distribution operate in coordination, irrigation water contributes more strongly to output. Villupuram illustrates this mechanism. Canal networks (845 km) operate alongside tank systems in sub-basins such as Gadilam, enabling water allocation across crop stages and alignment with complementary inputs. This results in higher marginal returns.

In contrast, districts with incomplete system integration exhibit lower economic water values. In Krishnagiri (137 km canal length) and Dharmapuri (187 km), reservoir storage does not translate into effective irrigation. Although the Krishnagiri reservoir has a storage capacity of 66.10 Mcum, limited canal extension restricts distribution. This reduces the contribution of irrigation water to production and shows that storage alone does not determine irrigation performance.

A similar pattern is observed in Cuddalore, where extensive canal length (1058 km) does not correspond to higher economic water values. Infrastructure expansion without flow regulation and field-level control does not improve irrigation performance. This indicates that infrastructure quality and operational management determine outcomes rather than scale alone.

The results also clarify the role of decentralised storage systems. In Thiruvannamalai, irrigation depends on tank systems concentrated in sub-basins such as Thurinjalar. These provide local storage under limited canal coverage (119 km), but lack of integration constrains irrigation benefits. Salem shows a comparable structure, where moderate canal coverage (233 km) combined with fragmented systems limits coordination across water sources.

Taken together, these findings show that irrigation water productivity depends on the interaction between storage, conveyance, and distribution systems. Infrastructure components operate jointly, and their combined performance determines how water contributes to production.

The estimated economic values represent marginal economic values (shadow prices) derived from production function estimates. These values reflect the contribution of water under observed input combinations and do not correspond to observed market prices or administratively determined tariffs.

The results provide a basis for reassessing national water policy. The National Water Policy (2012) applies area-based pricing without reference to marginal productivity. The estimated values show that water productivity differs across districts due to infrastructure conditions. This implies that uniform pricing does not reflect water's contribution and cannot support efficient allocation.

Across the Pennaiyar River Basin, wet land assessments range from Rs. 2.50 to Rs. 20.87 per acre, while dry land assessments remain below Rs. 4 per acre (Commission, 2017). These values are substantially lower than the estimated marginal value of irrigation water. Variation in charges across districts does not correspond to variation in water productivity. Pricing remains area-based rather than volumetric, indicating that charges are not linked to actual water use.

The results support a differentiated policy framework based on district-level conditions. In districts with integrated infrastructure, such as Villupuram and Vellore, irrigation authorities can introduce volumetric pricing through installation of flow measurement devices at canal outlets and groundwater extraction points. This can be combined with performance-linked incentives that relate water use to crop output.

In districts with incomplete systems, such as Dharmapuri and Krishnagiri, policy intervention should prioritise canal rehabilitation, tank system integration, and groundwater recharge structures to improve water delivery efficiency before pricing reforms are introduced. This establishes that pricing policy must follow infrastructure conditions within basin systems (National Water Mission, Government of India, 2017).

Implementation requires consideration of transaction costs. Extending pricing systems to volumetric formats requires investment in measurement infrastructure, monitoring systems, and administrative capacity. The absence of volumetric metering and reliance on area-based

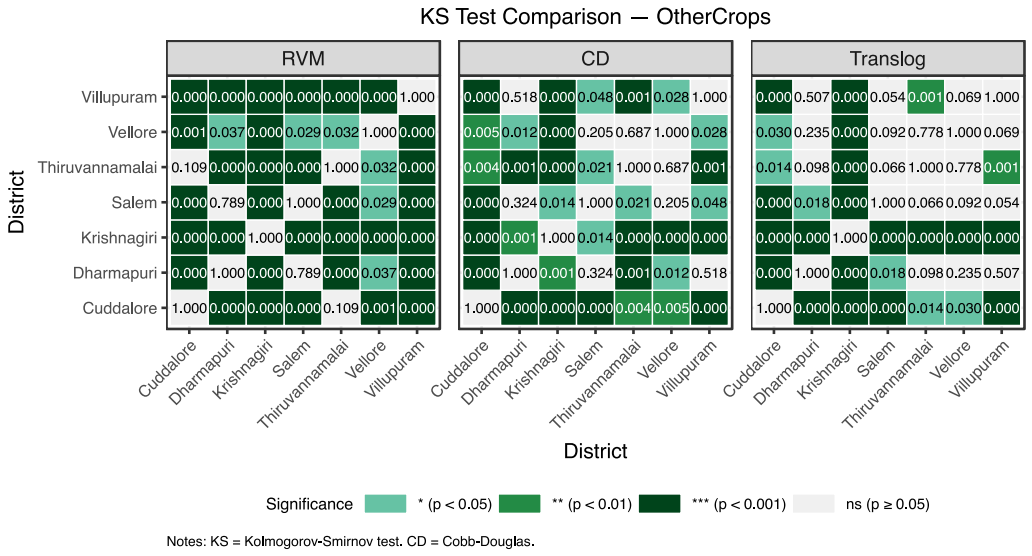


Fig. 7. Kolmogorov–Smirnov (KS) test heatmap for other crops, showing pairwise statistical comparisons of economic irrigation water value distributions across districts under the Residual Value Method (RVM), Cobb–Douglas (CD), and Translog production functions. Cell values represent KS test p-values, and colour intensity indicates the level of statistical significance between district-level distributions.

systems indicate these constraints. Spatial and temporal variability in water delivery increases implementation requirements. Adoption of pricing reforms depends on whether the expected efficiency gains exceed implementation costs.

The results extend existing evidence on irrigation and agricultural productivity. Previous studies link irrigation investment to agricultural growth (Fan et al., 2000; Rosegrant and Evenson, 1992; Hussain and Hanjra, 2004; Bhattarai et al., 2002). The present analysis shows that infrastructure investment alone does not determine outcomes. The interaction between infrastructure components determines irrigation performance. This shifts the focus from expansion to system configuration, consistent with work on water use efficiency (Molden et al., 2010).

The scope of the analysis is defined by data and model structure. Secondary data introduce measurement error in input and output values. The Translog specification requires larger sample sizes and is sensitive to multicollinearity. Groundwater extraction costs and environmental externalities are not included.

Future research can extend the present framework along three directions. First, incorporation of groundwater extraction behaviour, energy pricing, and climate variability can improve estimation of irrigation water value under changing resource conditions. Second, integration of spatial infrastructure data, including canal networks, tank systems, and groundwater availability, can strengthen analysis of how water distribution affects production outcomes. Third, combining production-based valuation with farm-level survey data can enable assessment of farmer behaviour, including input use decisions and willingness to pay for irrigation services. These extensions link economic valuation with hydrological and institutional conditions in irrigation systems.

6. Conclusion

This study estimates the economic value of irrigation water in the Pennaiyar River Basin using the Residual Value Method, Cobb–Douglas, and Translog production functions. Using crop-level data across five agricultural years and seven districts, the analysis quantifies the marginal contribution of irrigation water and examines its variation across space and time.

The results show that the economic value of irrigation water varies across production environments. This variation is determined by rainfall conditions, crop characteristics, and the coordination of complementary inputs. Higher marginal values are observed in settings where irrigation is combined with labour, machinery, and material inputs, while lower values correspond to conditions where rainfall reduces the marginal contribution of irrigation or where input use is not aligned. These findings establish that irrigation productivity depends on input coordination rather than water availability alone.

The Cobb–Douglas specification provides stable elasticity estimates and serves as a baseline for comparison, while the Translog specification captures variation in input relationships through interaction effects. The interaction terms represent how the contribution of one input depends on the level of another input. For example, a negative interaction between labour and machinery indicates that an increase in mechanisation reduces the contribution of labour to output, reflecting substitution between these inputs. A positive interaction between labour and material inputs indicates that joint use of these inputs increases their combined contribution to output, reflecting complementarity. Interaction terms involving irrigation water indicate that the contribution of water depends on the combination of other inputs used in production. These results show that irrigation productivity is determined by joint input use rather than by water alone.

These findings provide a basis for differentiated irrigation policy. In production environments with higher marginal values, policy should focus on improving water use efficiency through volumetric pricing, monitoring systems, and input coordination. In environments with lower marginal values, priority should be given to strengthening irrigation infrastructure, including canal systems, tank integration, and groundwater recharge, before pricing mechanisms are implemented. These findings contribute to the international literature by demonstrating that irrigation water valuation requires consideration of spatial variation and input coordination within production systems, which has implications for policy design in other water-scarce regions.

The estimated values represent marginal economic values derived from production relationships rather than observed prices. These estimates provide a benchmark for policy evaluation but do not represent

directly implementable tariffs. Implementation requires investment in volumetric measurement systems and institutional capacity.

The analysis is subject to limitations. The panel covers a limited time period, and the estimation relies on specific functional form assumptions, particularly in the Translog specification. These factors affect the stability and interpretation of estimated marginal values.

Future research can extend this framework by incorporating groundwater dynamics, infrastructure conditions, and farmer behaviour to improve irrigation water valuation.

CRediT authorship contribution statement

Sam Sundar Singh: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Parthasarathy Ramachandran:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Author Declaration

The authors used generative AI tools (ChatGPT by OpenAI) to assist in improving the language and readability of the manuscript. All content was reviewed and edited by the authors, who take full responsibility for the final version of the work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Regression tables

A.1.1. Cobb–Douglas estimates for paddy

See Table A.6.

A.1.2. Cobb–Douglas estimates for sugarcane

See Table A.7.

A.1.3. Cobb–Douglas estimates for other crops

See Table A.8.

A.1.4. Translog estimates for paddy

See Table A.9.

A.1.5. Translog estimates for sugarcane

See Table A.10.

A.1.6. Translog estimates for other crops

See Table A.11.

A.2. KS test: Significant district comparisons

See Table A.12.

Table A.6
Cobb–Douglas production function estimates for paddy.

	Dependent variable: Agricultural output				
	2017–18	2018–19	2019–20	2020–21	2021–22
Human labour	0.380***	0.255***	0.723***	0.411***	0.316**
	(0.063)	(0.053)	(0.050)	(0.107)	(0.140)
Animal labour	0.208***	−0.030	−0.007	−0.007	−0.062**
	(0.054)	(0.029)	(0.012)	(0.026)	(0.024)
Machinery	−0.119**	0.181**	0.055	0.157*	0.095
	(0.052)	(0.080)	(0.070)	(0.088)	(0.075)
Material input	0.118	0.070	−0.137**	0.227**	0.143**
	(0.098)	(0.078)	(0.064)	(0.102)	(0.072)
Water	0.051	0.076***	0.048***	0.197***	0.094***
	(0.032)	(0.022)	(0.019)	(0.042)	(0.036)
Constant	2.136***	0.878***	1.197***	−0.429	0.560
	(0.297)	(0.267)	(0.181)	(0.452)	(0.369)
Observations	207	174	242	213	216
Mean VIF	1.137	1.097	1.175	1.362	1.341
BP test (p-value)	0.070	0.004	0.004	0.000	0.006
RESET test (p-value)	0.049	0.522	0.068	0.003	0.402
DW test (p-value)	0.000	0.005	0.000	0.000	0.002
R^2	0.193	0.181	0.456	0.199	0.202
Adjusted R^2	0.173	0.157	0.444	0.180	0.183

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table A.7
Cobb–Douglas production function estimates for sugarcane.

	Dependent variable: Agricultural output			
	2017–18	2018–19	2019–20	2020–21
Human labour	0.437	0.338**	0.542***	0.597***
	(0.289)	(0.151)	(0.123)	(0.110)
Animal labour	0.062	−0.002	−0.115**	−0.023
	(0.092)	(0.038)	(0.049)	(0.025)
Machinery	0.065	0.057	−0.125**	0.105**
	(0.064)	(0.053)	(0.058)	(0.048)
Material input	0.103	0.058	−0.054	−0.063***
	(0.076)	(0.056)	(0.118)	(0.021)
Water	0.182*	0.023	0.029	0.022
	(0.104)	(0.034)	(0.055)	(0.026)
Constant	0.551	2.043***	1.578**	1.691***
	(1.358)	(0.472)	(0.663)	(0.409)
Observations	74	84	56	90
Mean VIF	1.415	1.485	1.243	1.387
BP test (p-value)	0.001	0.035	0.381	0.075
RESET test (p-value)	0.019	0.024	0.217	0.602
DW test (p-value)	0.509	0.042	0.002	0.000
R^2	0.275	0.427	0.377	0.486
Adjusted R^2	0.222	0.391	0.315	0.455

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

A.3. Data structure and sample distribution for paddy

See Table A.13.

A.4. Data structure and sample distribution for sugarcane

See Table A.14.

A.5. Data structure and sample distribution for other crops

See Table A.15.

Table A.8
Cobb–Douglas production function estimates for other crops.

	Dependent variable: Agricultural output				
	2017–18	2018–19	2019–20	2020–21	2021–22
Human labour	0.352***	0.566***	0.538***	0.446***	0.698***
	(0.091)	(0.077)	(0.064)	(0.063)	(0.114)
Animal labour	−0.015	−0.066***	0.071***	−0.025	0.024
	(0.022)	(0.023)	(0.013)	(0.017)	(0.015)
Machinery	0.167**	0.093*	0.125*	0.166***	0.181***
	(0.067)	(0.051)	(0.066)	(0.034)	(0.045)
Material input	0.238**	0.200***	0.007	0.377***	0.190**
	(0.103)	(0.077)	(0.038)	(0.056)	(0.079)
Water	0.203***	0.063*	0.074**	0.215***	0.178***
	(0.074)	(0.036)	(0.033)	(0.033)	(0.053)
Constant	−0.182	0.533*	1.174***	−0.347	−0.010
	(0.610)	(0.323)	(0.279)	(0.242)	(0.423)
Observations	189	220	247	199	187
Mean VIF	1.380	1.488	1.326	2.185	2.792
BP test (*p*-value)	0.001	0.259	0.102	0.118	0.009
RESET test (*p*-value)	0.505	0.003	0.013	0.000	0.002
DW test (*p*-value)	0.000	0.000	0.000	0.000	0.000
R^2	0.389	0.506	0.463	0.833	0.755
Adjusted R^2	0.372	0.495	0.452	0.828	0.749

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table A.9
Translog production function estimates for paddy.

	Dependent variable: Agricultural output				
	2017–18	2018–19	2019–20	2020–21	2021-22
Human labour	0.896	3.370***	−2.159**	2.221	3.282
	(1.642)	(1.219)	(0.891)	(2.307)	(3.075)
Animal labour	2.099*	1.410***	0.092	0.879**	−0.219
	(1.136)	(0.520)	(0.296)	(0.423)	(0.659)
Machinery	−2.114**	0.982	−1.979**	1.705	−0.884
	(0.902)	(1.481)	(0.858)	(1.473)	(1.063)
Material inputs	−2.930	−2.488	−0.264	0.812	−0.208
	(2.193)	(1.827)	(0.973)	(1.521)	(1.540)
Water	0.694	−0.066	−0.192	2.127**	−0.169
	(0.692)	(0.500)	(0.377)	(1.003)	(0.914)
Human labour^2	−0.464	0.116	0.023	−0.212	−1.674
	(0.316)	(0.152)	(0.288)	(0.715)	(1.050)
Animal labour^2	−0.381	−0.241	0.060	0.047	−0.240
	(0.335)	(0.194)	(0.047)	(0.096)	(0.166)
Machinery^2	−0.057	0.343	−0.442*	−0.397	0.285
	(0.138)	(0.326)	(0.265)	(0.591)	(0.221)
Material inputs^2	−0.234	1.357**	−0.442	−0.856	−0.023
	(0.490)	(0.600)	(0.427)	(0.573)	(0.274)
Water^2	−0.163*	−0.038	0.006	−0.202**	0.009
	(0.098)	(0.040)	(0.039)	(0.097)	(0.091)
Human × Animal	−0.224	−0.059	−0.241***	−0.092	−0.161
	(0.224)	(0.152)	(0.087)	(0.108)	(0.141)
Human × Machinery	0.111	−0.763***	−0.468**	0.167	0.447
	(0.191)	(0.266)	(0.228)	(0.460)	(0.349)
Human × Material inputs	0.342	−0.635**	0.709**	−0.250	0.522
	(0.308)	(0.300)	(0.312)	(0.299)	(0.425)
Human × Water	−0.137	−0.282***	0.196**	−0.228	−0.353
	(0.135)	(0.085)	(0.082)	(0.217)	(0.318)
Animal × Machinery	−0.381	−0.228*	−0.176**	−0.067	−0.065
	(0.232)	(0.126)	(0.078)	(0.109)	(0.169)
Animal × Material inputs	−1.062***	−0.388	0.079	−0.264**	0.190*
	(0.375)	(0.239)	(0.076)	(0.126)	(0.108)
Animal × Water	−0.261***	−0.182***	0.032	−0.053	−0.067
	(0.082)	(0.041)	(0.024)	(0.043)	(0.047)
Machinery × Material inputs	−0.152	0.033	−0.352	−0.372	−0.522***
	(0.324)	(0.291)	(0.316)	(0.318)	(0.188)
Machinery × Water	0.056	−0.158	0.223**	−0.158	0.045
	(0.121)	(0.117)	(0.087)	(0.144)	(0.092)

(*continued on next page*)

Table A.9 (*continued*).

	Dependent variable: Agricultural output				
	2017–18	2018–19	2019–20	2020–21	2021-22
Material inputs × Water	−0.143	0.060	0.055	−0.082	0.140
	(0.158)	(0.120)	(0.091)	(0.140)	(0.154)
Constant	4.190	4.680	4.072**	−8.751*	1.573
	(3.747)	(3.446)	(2.072)	(5.274)	(4.896)
Observations	207	174	242	213	216
BP Test (*p*-value)	0.135	0.093	0.005	0.019	0.224
R^2	0.251	0.32	0.534	0.274	0.264
Adjusted R^2	0.171	0.231	0.492	0.198	0.189

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table A.10
Translog production function estimates for sugarcane.

	Dependent variable: Agricultural output			
	2017–18	2018–19	2019–20	2020-21
Human labour	1.733	1.520	−1.645	−0.615
	(4.399)	(1.538)	(2.299)	(2.071)
Animal labour	1.538	−1.593	0.924	−0.070
	(2.131)	(1.013)	(1.815)	(0.296)
Machinery	0.639	0.509	−2.606	0.218
	(1.780)	(0.678)	(1.737)	(0.702)
Material inputs	4.226**	3.230***	−1.078	1.157***
	(2.045)	(0.977)	(1.818)	(0.442)
Water	1.392	1.807**	1.012	−1.965***
	(1.929)	(0.813)	(1.048)	(0.504)
Human labour^2	0.375	0.696***	0.884**	−0.791
	(0.976)	(0.191)	(0.366)	(0.739)
Animal labour^2	0.051	0.062	−0.186	0.021
	(0.433)	(0.090)	(0.180)	(0.064)
Machinery^2	−0.120	0.084	−0.081	−0.143
	(0.192)	(0.075)	(0.190)	(0.098)
Material inputs^2	−1.277**	0.086	−0.396	0.046
	(0.638)	(0.125)	(0.320)	(0.041)
Water^2	−0.147	−0.147**	−0.074	0.144***
	(0.186)	(0.072)	(0.111)	(0.049)
Human × Animal	0.479	0.562*	−0.670	0.004
	(0.615)	(0.315)	(0.557)	(0.089)
Human × Machinery	0.252	−0.132	0.675***	0.233
	(0.479)	(0.095)	(0.257)	(0.189)
Human × Material inputs	−0.721	−0.556**	0.060	−0.280*
	(0.573)	(0.230)	(0.326)	(0.160)
Human × Water	−0.064	0.005	−0.157	0.319***
	(0.263)	(0.108)	(0.199)	(0.097)
Animal × Machinery	−0.060	0.121**	0.009	−0.088***
	(0.152)	(0.061)	(0.124)	(0.027)
Animal × Material inputs	−0.046	−0.004	−0.008	−0.019
	(0.271)	(0.064)	(0.166)	(0.018)
Animal × Water	−0.254	0.044	0.016	0.010
	(0.220)	(0.052)	(0.088)	(0.023)
Machinery × Material inputs	−0.022	0.162**	0.001	−0.082
	(0.128)	(0.074)	(0.178)	(0.059)
Machinery × Water	−0.122	−0.024	0.123	−0.069
	(0.127)	(0.061)	(0.132)	(0.045)
Material inputs × Water	−0.180	−0.219***	0.119	−0.068**
	(0.143)	(0.074)	(0.165)	(0.028)
Constant	−6.656	−12.610**	1.161	11.982***
	(11.383)	(5.662)	(5.604)	(3.122)
Observations	74	84	56	90
BP test (*p*-value)	0.18	0.792	0.379	0.551
R^2	0.51	0.702	0.674	0.743
Adjusted R^2	0.325	0.607	0.487	0.669

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table A.11
Translog production function estimates for other crops.

	Dependent variable: Agricultural output				
	2017–18 (1)	2018–19 (2)	2019–20 (3)	2020–21 (4)	2021-22 (5)
Human labour	−2.563***	0.365	1.957**	−1.299*	1.349
	(0.742)	(0.688)	(0.922)	(0.761)	(1.465)
Animal labour	0.489*	−0.176	0.365	0.386	0.344
	(0.264)	(0.353)	(0.240)	(0.299)	(0.284)
Machinery	1.461**	1.159**	−0.220	0.212	0.686
	(0.668)	(0.555)	(0.719)	(0.441)	(0.598)
Material inputs	0.016	1.312**	−0.337	0.299	−1.330
	(0.714)	(0.548)	(0.624)	(0.561)	(0.958)
Water	−2.218***	−0.168	−0.142	0.683*	1.412**
	(0.506)	(0.515)	(0.392)	(0.388)	(0.690)
Human labour^2	0.712***	0.035	−0.895**	0.083	−0.229
	(0.199)	(0.235)	(0.376)	(0.371)	(0.480)
Animal labour^2	−0.013	0.016	−0.022	−0.006	−0.060***
	(0.018)	(0.025)	(0.016)	(0.025)	(0.022)
Machinery^2	0.167***	0.169*	0.225	0.099	0.475**
	(0.062)	(0.090)	(0.314)	(0.112)	(0.213)
Material inputs^2	−0.028	0.310*	0.038	0.055	0.134
	(0.228)	(0.187)	(0.053)	(0.161)	(0.300)
Water^2	0.216***	0.023	0.031	−0.107*	−0.168*
	(0.061)	(0.055)	(0.046)	(0.058)	(0.089)
Human × Animal	−0.022	0.107	−0.193**	−0.073	−0.289*
	(0.043)	(0.072)	(0.076)	(0.068)	(0.154)
Human × Machinery	−0.373***	−0.124	0.211	−0.252*	−0.838**
	(0.100)	(0.154)	(0.208)	(0.144)	(0.371)
Human × Material inputs	−0.428***	−0.263	0.342**	0.357*	0.362
	(0.130)	(0.175)	(0.142)	(0.214)	(0.283)
Human × Water	0.256***	0.066	−0.189*	0.162*	−0.149
	(0.079)	(0.075)	(0.112)	(0.093)	(0.194)
Animal × Machinery	0.065*	0.055	0.061	−0.001	−0.120**
	(0.037)	(0.051)	(0.039)	(0.037)	(0.061)
Animal × Material inputs	0.078	0.006	0.076**	0.124**	0.203***
	(0.069)	(0.063)	(0.038)	(0.058)	(0.073)
Animal × Water	−0.068**	0.011	−0.030	−0.049	−0.035
	(0.033)	(0.037)	(0.024)	(0.033)	(0.030)
Machinery × Material inputs	−0.002	−0.086	−0.353*	−0.025	0.011
	(0.112)	(0.091)	(0.201)	(0.115)	(0.210)
Machinery × Water	−0.069	−0.083	0.065	0.031	−0.006
	(0.072)	(0.062)	(0.090)	(0.052)	(0.073)
Material inputs × Water	0.135	−0.099	0.041	0.034	0.214*
	(0.086)	(0.064)	(0.075)	(0.077)	(0.118)
Constant	12.482***	1.405	2.073	−0.574	−4.313
	(2.284)	(2.645)	(1.800)	(1.443)	(2.862)
Observations	189	220	247	199	187
BP Test (*p*-value)	0.325	0.105	0	0.915	0.004
R^2	0.637	0.576	0.524	0.867	0.799
Adjusted R^2	0.594	0.533	0.482	0.853	0.775

Note: Robust standard errors (HC1) in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table A.12
KS test: Significant district comparisons by Year ($p < 0.05$).

Crop	Model	Year	District 1	District 2	D-statistic
Paddy	RVM	2017–18	Cuddalore	Krishnagiri	0.73
Paddy	RVM	2017–18	Cuddalore	Thiruvannamalai	0.33
Paddy	RVM	2017–18	Cuddalore	Vellore	0.57
Paddy	RVM	2017–18	Cuddalore	Villupuram	0.50
Paddy	RVM	2017–18	Dharmapuri	Villupuram	0.53
Paddy	RVM	2017–18	Krishnagiri	Thiruvannamalai	0.59
Paddy	RVM	2017–18	Krishnagiri	Villupuram	0.88
Paddy	RVM	2017–18	Salem	Villupuram	1.00
Paddy	RVM	2017–18	Thiruvannamalai	Villupuram	0.77
Paddy	RVM	2017–18	Vellore	Villupuram	0.71
Paddy	RVM	2018–19	Cuddalore	Krishnagiri	0.83
Paddy	RVM	2018–19	Cuddalore	Thiruvannamalai	0.38
Paddy	RVM	2018–19	Cuddalore	Vellore	0.91
Paddy	RVM	2018–19	Cuddalore	Villupuram	0.54
Paddy	RVM	2018–19	Krishnagiri	Thiruvannamalai	0.68
Paddy	RVM	2018–19	Krishnagiri	Villupuram	0.76
Paddy	RVM	2018–19	Vellore	Villupuram	1.00
Paddy	RVM	2019–20	Cuddalore	Dharmapuri	0.56
Paddy	RVM	2019–20	Cuddalore	Krishnagiri	0.92
Paddy	RVM	2019–20	Cuddalore	Salem	0.90
Paddy	RVM	2019–20	Cuddalore	Vellore	0.73
Paddy	RVM	2019–20	Cuddalore	Villupuram	0.59
Paddy	RVM	2019–20	Dharmapuri	Krishnagiri	0.83
Paddy	RVM	2019–20	Dharmapuri	Thiruvannamalai	0.55
Paddy	RVM	2019–20	Dharmapuri	Vellore	0.83
Paddy	RVM	2019–20	Dharmapuri	Villupuram	0.83
Paddy	RVM	2019–20	Krishnagiri	Thiruvannamalai	0.88
Paddy	RVM	2019–20	Krishnagiri	Vellore	1.00
Paddy	RVM	2019–20	Krishnagiri	Villupuram	1.00
Paddy	RVM	2019–20	Salem	Villupuram	1.00
Paddy	RVM	2019–20	Thiruvannamalai	Vellore	0.70
Paddy	RVM	2019–20	Thiruvannamalai	Villupuram	0.61
Paddy	RVM	2019–20	Vellore	Villupuram	0.69
Paddy	RVM	2020–21	Cuddalore	Thiruvannamalai	0.35
Paddy	RVM	2020–21	Cuddalore	Vellore	0.29
Paddy	RVM	2020–21	Cuddalore	Villupuram	0.30
Paddy	RVM	2020–21	Dharmapuri	Villupuram	0.67
Paddy	RVM	2020–21	Thiruvannamalai	Vellore	0.39
Paddy	RVM	2020–21	Vellore	Villupuram	0.52
Paddy	RVM	2021–22	Cuddalore	Salem	0.65
Paddy	RVM	2021–22	Cuddalore	Vellore	0.39
Paddy	RVM	2021–22	Salem	Thiruvannamalai	0.62
Paddy	RVM	2021–22	Salem	Villupuram	0.68
Paddy	RVM	2021–22	Thiruvannamalai	Vellore	0.37
Paddy	RVM	2021–22	Vellore	Villupuram	0.44
Paddy	CD	2017–18	Cuddalore	Dharmapuri	0.39
Paddy	CD	2017–18	Cuddalore	Krishnagiri	0.68
Paddy	CD	2017–18	Cuddalore	Salem	0.83
Paddy	CD	2017–18	Cuddalore	Thiruvannamalai	0.23
Paddy	CD	2017–18	Cuddalore	Villupuram	0.49
Paddy	CD	2017–18	Dharmapuri	Villupuram	0.76
Paddy	CD	2017–18	Krishnagiri	Thiruvannamalai	0.72
Paddy	CD	2017–18	Krishnagiri	Villupuram	0.88
Paddy	CD	2017–18	Salem	Thiruvannamalai	0.89
Paddy	CD	2017–18	Salem	Villupuram	1.00
Paddy	CD	2017–18	Thiruvannamalai	Villupuram	0.69
Paddy	CD	2017–18	Vellore	Villupuram	0.68
Paddy	CD	2018–19	Cuddalore	Dharmapuri	0.81
Paddy	CD	2018–19	Cuddalore	Krishnagiri	0.75
Paddy	CD	2018–19	Cuddalore	Thiruvannamalai	0.26
Paddy	CD	2018–19	Dharmapuri	Villupuram	0.89
Paddy	CD	2018–19	Krishnagiri	Thiruvannamalai	0.57
Paddy	CD	2018–19	Krishnagiri	Vellore	1.00
Paddy	CD	2018–19	Krishnagiri	Villupuram	0.89
Paddy	CD	2019–20	Cuddalore	Krishnagiri	0.76
Paddy	CD	2019–20	Cuddalore	Vellore	0.74
Paddy	CD	2019–20	Cuddalore	Villupuram	0.45
Paddy	CD	2019–20	Dharmapuri	Villupuram	0.78
Paddy	CD	2019–20	Krishnagiri	Thiruvannamalai	0.73
Paddy	CD	2019–20	Krishnagiri	Vellore	0.78
Paddy	CD	2019–20	Krishnagiri	Villupuram	0.94
Paddy	CD	2019–20	Salem	Villupuram	0.94
Paddy	CD	2019–20	Thiruvannamalai	Vellore	0.72
Paddy	CD	2019–20	Thiruvannamalai	Villupuram	0.56
Paddy	CD	2019–20	Vellore	Villupuram	0.94
Paddy	CD	2020–21	Cuddalore	Dharmapuri	0.70
Paddy	CD	2020–21	Cuddalore	Salem	0.62
Paddy	CD	2020–21	Dharmapuri	Thiruvannamalai	0.66
Paddy	CD	2020–21	Dharmapuri	Vellore	0.57
Paddy	CD	2020–21	Dharmapuri	Villupuram	0.73
Paddy	CD	2020–21	Salem	Thiruvannamalai	0.66
Paddy	CD	2020–21	Salem	Vellore	0.49
Paddy	CD	2020–21	Salem	Villupuram	0.64
Paddy	CD	2021–22	Cuddalore	Salem	0.56

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Table A.12 (*continued*).

Crop	Model	Year	District 1	District 2	D-statistic
Paddy	CD	2021–22	Cuddalore	Vellore	0.28
Paddy	CD	2021–22	Salem	Thiruvannamalai	0.58
Paddy	CD	2021–22	Salem	Vellore	0.70
Paddy	CD	2021–22	Salem	Villupuram	0.59
Paddy	CD	2021–22	Thiruvannamalai	Vellore	0.36
Paddy	Translog	2017–18	Cuddalore	Dharmapuri	0.43
Paddy	Translog	2017–18	Cuddalore	Krishnagiri	0.68
Paddy	Translog	2017–18	Cuddalore	Salem	0.86
Paddy	Translog	2017–18	Cuddalore	Thiruvannamalai	0.24
Paddy	Translog	2017–18	Cuddalore	Villupuram	0.46
Paddy	Translog	2017–18	Dharmapuri	Thiruvannamalai	0.46
Paddy	Translog	2017–18	Dharmapuri	Villupuram	0.76
Paddy	Translog	2017–18	Krishnagiri	Thiruvannamalai	0.69
Paddy	Translog	2017–18	Krishnagiri	Villupuram	0.82
Paddy	Translog	2017–18	Salem	Thiruvannamalai	0.92
Paddy	Translog	2017–18	Salem	Villupuram	1.00
Paddy	Translog	2017–18	Thiruvannamalai	Vellore	0.51
Paddy	Translog	2017–18	Thiruvannamalai	Villupuram	0.69
Paddy	Translog	2017–18	Vellore	Villupuram	0.68
Paddy	Translog	2018–19	Cuddalore	Dharmapuri	0.96
Paddy	Translog	2018–19	Cuddalore	Thiruvannamalai	0.65
Paddy	Translog	2018–19	Cuddalore	Villupuram	0.79
Paddy	Translog	2018–19	Thiruvannamalai	Villupuram	0.46
Paddy	Translog	2019–20	Cuddalore	Dharmapuri	0.58
Paddy	Translog	2019–20	Cuddalore	Krishnagiri	0.76
Paddy	Translog	2019–20	Cuddalore	Vellore	0.74
Paddy	Translog	2019–20	Cuddalore	Villupuram	0.34
Paddy	Translog	2019–20	Dharmapuri	Villupuram	0.83
Paddy	Translog	2019–20	Krishnagiri	Thiruvannamalai	0.71
Paddy	Translog	2019–20	Krishnagiri	Vellore	0.78
Paddy	Translog	2019–20	Krishnagiri	Villupuram	0.94
Paddy	Translog	2019–20	Salem	Villupuram	0.94
Paddy	Translog	2019–20	Thiruvannamalai	Vellore	0.66
Paddy	Translog	2019–20	Thiruvannamalai	Villupuram	0.41
Paddy	Translog	2019–20	Vellore	Villupuram	0.94
Paddy	Translog	2020–21	Cuddalore	Dharmapuri	0.59
Paddy	Translog	2020–21	Cuddalore	Salem	0.53
Paddy	Translog	2020–21	Cuddalore	Thiruvannamalai	0.31
Paddy	Translog	2020–21	Cuddalore	Vellore	0.32
Paddy	Translog	2020–21	Salem	Thiruvannamalai	0.48
Paddy	Translog	2020–21	Salem	Villupuram	0.49
Paddy	Translog	2021–22	Thiruvannamalai	Vellore	0.26
Sugarcane	RVM	2017–18	Cuddalore	Vellore	1.00
Sugarcane	RVM	2017–18	Thiruvannamalai	Vellore	0.96
Sugarcane	RVM	2017–18	Vellore	Villupuram	1.00
Sugarcane	RVM	2018–19	Cuddalore	Vellore	0.87
Sugarcane	RVM	2018–19	Cuddalore	Villupuram	0.51
Sugarcane	RVM	2018–19	Krishnagiri	Vellore	1.00
Sugarcane	RVM	2018–19	Krishnagiri	Villupuram	0.69
Sugarcane	RVM	2018–19	Thiruvannamalai	Vellore	0.74
Sugarcane	RVM	2018–19	Thiruvannamalai	Villupuram	0.59
Sugarcane	RVM	2019–20	Krishnagiri	Villupuram	0.80
Sugarcane	RVM	2020–21	Cuddalore	Dharmapuri	0.38
Sugarcane	RVM	2020–21	Dharmapuri	Thiruvannamalai	0.60
Sugarcane	RVM	2020–21	Dharmapuri	Vellore	0.71
Sugarcane	CD	2018–19	Cuddalore	Thiruvannamalai	0.40
Sugarcane	CD	2018–19	Cuddalore	Villupuram	0.56
Sugarcane	CD	2018–19	Dharmapuri	Thiruvannamalai	0.90
Sugarcane	CD	2018–19	Dharmapuri	Villupuram	1.00
Sugarcane	CD	2018–19	Krishnagiri	Thiruvannamalai	0.52
Sugarcane	CD	2018–19	Krishnagiri	Villupuram	0.67
Sugarcane	CD	2020–21	Cuddalore	Thiruvannamalai	0.41
Sugarcane	CD	2020–21	Dharmapuri	Thiruvannamalai	0.62
Sugarcane	CD	2020–21	Dharmapuri	Vellore	0.66
Sugarcane	Translog	2018–19	Cuddalore	Villupuram	0.46
Sugarcane	Translog	2018–19	Dharmapuri	Thiruvannamalai	0.84
Sugarcane	Translog	2018–19	Dharmapuri	Villupuram	1.00
Sugarcane	Translog	2018–19	Krishnagiri	Villupuram	0.62
Sugarcane	Translog	2019–20	Cuddalore	Krishnagiri	0.56
Sugarcane	Translog	2020–21	Cuddalore	Vellore	0.58
Sugarcane	Translog	2020–21	Dharmapuri	Thiruvannamalai	0.54
Sugarcane	Translog	2020–21	Dharmapuri	Vellore	0.69

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Table A.12 (*continued*).

Crop	Model	Year	District 1	District 2	D-statistic
Other crops	RVM	2017–18	Cuddalore	Dharmapuri	0.55
Other crops	RVM	2017–18	Cuddalore	Krishnagiri	0.48
Other crops	RVM	2017–18	Cuddalore	Salem	0.56
Other crops	RVM	2017–18	Cuddalore	Thiruvannamalai	0.35
Other crops	RVM	2017–18	Cuddalore	Vellore	0.71
Other crops	RVM	2017–18	Cuddalore	Villupuram	0.53
Other crops	RVM	2017–18	Dharmapuri	Thiruvannamalai	0.62
Other crops	RVM	2017–18	Dharmapuri	Vellore	0.71
Other crops	RVM	2017–18	Dharmapuri	Villupuram	0.72
Other crops	RVM	2017–18	Krishnagiri	Thiruvannamalai	0.55
Other crops	RVM	2017–18	Krishnagiri	Vellore	0.54
Other crops	RVM	2017–18	Krishnagiri	Villupuram	0.55
Other crops	RVM	2017–18	Salem	Thiruvannamalai	0.60
Other crops	RVM	2017–18	Salem	Vellore	0.61
Other crops	RVM	2017–18	Salem	Villupuram	0.67
Other crops	RVM	2017–18	Thiruvannamalai	Vellore	0.51
Other crops	RVM	2017–18	Vellore	Villupuram	0.44
Other crops	RVM	2018–19	Cuddalore	Dharmapuri	0.38
Other crops	RVM	2018–19	Cuddalore	Salem	0.62
Other crops	RVM	2018–19	Cuddalore	Thiruvannamalai	0.37
Other crops	RVM	2018–19	Cuddalore	Vellore	0.39
Other crops	RVM	2018–19	Cuddalore	Villupuram	0.62
Other crops	RVM	2018–19	Dharmapuri	Salem	0.53
Other crops	RVM	2018–19	Dharmapuri	Vellore	0.49
Other crops	RVM	2018–19	Dharmapuri	Villupuram	0.56
Other crops	RVM	2018–19	Salem	Thiruvannamalai	0.44
Other crops	RVM	2018–19	Salem	Villupuram	0.78
Other crops	RVM	2018–19	Thiruvannamalai	Villupuram	0.57
Other crops	RVM	2018–19	Vellore	Villupuram	0.71
Other crops	RVM	2019–20	Cuddalore	Dharmapuri	0.44
Other crops	RVM	2019–20	Cuddalore	Salem	0.44
Other crops	RVM	2019–20	Cuddalore	Thiruvannamalai	0.41
Other crops	RVM	2019–20	Cuddalore	Villupuram	0.62
Other crops	RVM	2019–20	Dharmapuri	Thiruvannamalai	0.36
Other crops	RVM	2019–20	Dharmapuri	Vellore	0.34
Other crops	RVM	2019–20	Dharmapuri	Villupuram	0.51
Other crops	RVM	2019–20	Krishnagiri	Villupuram	0.71
Other crops	RVM	2019–20	Salem	Villupuram	0.63
Other crops	RVM	2019–20	Thiruvannamalai	Villupuram	0.84
Other crops	RVM	2019–20	Vellore	Villupuram	0.80
Other crops	RVM	2020–21	Cuddalore	Krishnagiri	0.69
Other crops	RVM	2020–21	Cuddalore	Salem	0.46
Other crops	RVM	2020–21	Cuddalore	Thiruvannamalai	0.36
Other crops	RVM	2020–21	Cuddalore	Vellore	0.55
Other crops	RVM	2020–21	Dharmapuri	Krishnagiri	0.66
Other crops	RVM	2020–21	Krishnagiri	Salem	0.75
Other crops	RVM	2020–21	Krishnagiri	Thiruvannamalai	0.71
Other crops	RVM	2020–21	Krishnagiri	Vellore	0.86
Other crops	RVM	2020–21	Krishnagiri	Villupuram	0.74
Other crops	RVM	2020–21	Salem	Thiruvannamalai	0.52
Other crops	RVM	2020–21	Thiruvannamalai	Vellore	0.85
Other crops	RVM	2020–21	Vellore	Villupuram	0.62
Other crops	RVM	2021–22	Cuddalore	Krishnagiri	0.59
Other crops	RVM	2021–22	Cuddalore	Salem	0.57
Other crops	RVM	2021–22	Cuddalore	Thiruvannamalai	0.33
Other crops	RVM	2021–22	Cuddalore	Vellore	0.43
Other crops	RVM	2021–22	Cuddalore	Villupuram	0.61
Other crops	RVM	2021–22	Krishnagiri	Thiruvannamalai	0.65
Other crops	RVM	2021–22	Krishnagiri	Vellore	0.54
Other crops	RVM	2021–22	Salem	Thiruvannamalai	0.62
Other crops	RVM	2021–22	Thiruvannamalai	Vellore	0.41
Other crops	RVM	2021–22	Thiruvannamalai	Villupuram	0.67
Other crops	RVM	2021–22	Vellore	Villupuram	0.45
Other crops	CD	2017–18	Cuddalore	Dharmapuri	0.69
Other crops	CD	2017–18	Cuddalore	Salem	0.72
Other crops	CD	2017–18	Cuddalore	Thiruvannamalai	0.64
Other crops	CD	2017–18	Cuddalore	Vellore	0.81
Other crops	CD	2017–18	Cuddalore	Villupuram	0.86
Other crops	CD	2017–18	Dharmapuri	Vellore	0.49
Other crops	CD	2017–18	Salem	Vellore	0.55
Other crops	CD	2017–18	Thiruvannamalai	Vellore	0.61
Other crops	CD	2017–18	Thiruvannamalai	Villupuram	0.47

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Table A.12 (*continued*).

Crop	Model	Year	District 1	District 2	D-statistic
Other crops	CD	2017–18	Vellore	Villupuram	0.45
Other crops	CD	2018–19	Cuddalore	Dharmapuri	0.67
Other crops	CD	2018–19	Cuddalore	Krishnagiri	0.97
Other crops	CD	2018–19	Cuddalore	Salem	0.40
Other crops	CD	2018–19	Cuddalore	Thiruvannamalai	0.29
Other crops	CD	2018–19	Cuddalore	Vellore	0.38
Other crops	CD	2018–19	Cuddalore	Villupuram	0.66
Other crops	CD	2018–19	Dharmapuri	Thiruvannamalai	0.61
Other crops	CD	2018–19	Krishnagiri	Thiruvannamalai	1.00
Other crops	CD	2018–19	Krishnagiri	Vellore	1.00
Other crops	CD	2018–19	Krishnagiri	Villupuram	0.94
Other crops	CD	2018–19	Salem	Villupuram	0.43
Other crops	CD	2018–19	Thiruvannamalai	Villupuram	0.59
Other crops	CD	2019–20	Cuddalore	Dharmapuri	0.74
Other crops	CD	2019–20	Cuddalore	Krishnagiri	0.88
Other crops	CD	2019–20	Cuddalore	Salem	0.64
Other crops	CD	2019–20	Cuddalore	Vellore	0.38
Other crops	CD	2019–20	Cuddalore	Villupuram	0.66
Other crops	CD	2019–20	Dharmapuri	Thiruvannamalai	0.70
Other crops	CD	2019–20	Dharmapuri	Vellore	0.60
Other crops	CD	2019–20	Krishnagiri	Thiruvannamalai	1.00
Other crops	CD	2019–20	Krishnagiri	Villupuram	0.93
Other crops	CD	2019–20	Salem	Thiruvannamalai	0.51
Other crops	CD	2019–20	Thiruvannamalai	Villupuram	0.50
Other crops	CD	2019–20	Vellore	Villupuram	0.38
Other crops	CD	2020–21	Cuddalore	Krishnagiri	0.51
Other crops	CD	2020–21	Cuddalore	Salem	0.68
Other crops	CD	2020–21	Cuddalore	Thiruvannamalai	0.31
Other crops	CD	2020–21	Krishnagiri	Thiruvannamalai	0.48
Other crops	CD	2020–21	Krishnagiri	Vellore	0.58
Other crops	CD	2020–21	Salem	Thiruvannamalai	0.66
Other crops	CD	2020–21	Salem	Vellore	0.82
Other crops	CD	2020–21	Salem	Villupuram	0.56
Other crops	CD	2021–22	Cuddalore	Salem	0.70
Other crops	CD	2021–22	Cuddalore	Vellore	0.35
Other crops	CD	2021–22	Cuddalore	Villupuram	0.54
Other crops	CD	2021–22	Salem	Thiruvannamalai	0.58
Other crops	CD	2021–22	Salem	Vellore	0.59
Other crops	CD	2021–22	Salem	Villupuram	0.53
Other crops	CD	2021–22	Thiruvannamalai	Villupuram	0.46
Other crops	CD	2021–22	Vellore	Villupuram	0.47
Other crops	Translog	2017–18	Cuddalore	Dharmapuri	0.38
Other crops	Translog	2017–18	Cuddalore	Salem	0.48
Other crops	Translog	2017–18	Cuddalore	Thiruvannamalai	0.64
Other crops	Translog	2017–18	Cuddalore	Vellore	0.74
Other crops	Translog	2017–18	Cuddalore	Villupuram	0.75
Other crops	Translog	2017–18	Dharmapuri	Thiruvannamalai	0.43
Other crops	Translog	2017–18	Dharmapuri	Vellore	0.60
Other crops	Translog	2017–18	Dharmapuri	Villupuram	0.49
Other crops	Translog	2017–18	Krishnagiri	Thiruvannamalai	0.47
Other crops	Translog	2017–18	Krishnagiri	Vellore	0.55
Other crops	Translog	2017–18	Krishnagiri	Villupuram	0.58
Other crops	Translog	2017–18	Salem	Vellore	0.50
Other crops	Translog	2017–18	Thiruvannamalai	Vellore	0.39
Other crops	Translog	2017–18	Vellore	Villupuram	0.46
Other crops	Translog	2018–19	Cuddalore	Dharmapuri	0.49
Other crops	Translog	2018–19	Cuddalore	Thiruvannamalai	0.31
Other crops	Translog	2018–19	Cuddalore	Villupuram	0.57
Other crops	Translog	2018–19	Dharmapuri	Vellore	0.53
Other crops	Translog	2018–19	Salem	Villupuram	0.52
Other crops	Translog	2018–19	Thiruvannamalai	Vellore	0.35
Other crops	Translog	2018–19	Thiruvannamalai	Villupuram	0.43
Other crops	Translog	2018–19	Vellore	Villupuram	0.54
Other crops	Translog	2019–20	Cuddalore	Dharmapuri	0.51
Other crops	Translog	2019–20	Cuddalore	Krishnagiri	0.80
Other crops	Translog	2019–20	Cuddalore	Thiruvannamalai	0.31
Other crops	Translog	2019–20	Cuddalore	Vellore	0.43
Other crops	Translog	2019–20	Cuddalore	Villupuram	0.24
Other crops	Translog	2019–20	Dharmapuri	Salem	0.45
Other crops	Translog	2019–20	Dharmapuri	Thiruvannamalai	0.67
Other crops	Translog	2019–20	Dharmapuri	Villupuram	0.40
Other crops	Translog	2019–20	Krishnagiri	Thiruvannamalai	0.95

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Table A.12 (*continued*).

Crop	Model	Year	District 1	District 2	D-statistic
Other crops	Translog	2019–20	Krishnagiri	Villupuram	0.68
Other crops	Translog	2019–20	Salem	Thiruvannamalai	0.38
Other crops	Translog	2019–20	Salem	Vellore	0.42
Other crops	Translog	2019–20	Thiruvannamalai	Vellore	0.42
Other crops	Translog	2019–20	Thiruvannamalai	Villupuram	0.35
Other crops	Translog	2019–20	Vellore	Villupuram	0.33
Other crops	Translog	2020–21	Cuddalore	Krishnagiri	0.70
Other crops	Translog	2020–21	Cuddalore	Salem	0.83
Other crops	Translog	2020–21	Cuddalore	Thiruvannamalai	0.35
Other crops	Translog	2020–21	Krishnagiri	Thiruvannamalai	0.75
Other crops	Translog	2020–21	Krishnagiri	Vellore	0.86
Other crops	Translog	2020–21	Krishnagiri	Villupuram	0.74
Other crops	Translog	2020–21	Salem	Thiruvannamalai	0.87
Other crops	Translog	2020–21	Salem	Vellore	0.91
Other crops	Translog	2020–21	Salem	Villupuram	0.81
Other crops	Translog	2021–22	Cuddalore	Krishnagiri	0.52
Other crops	Translog	2021–22	Cuddalore	Salem	0.74
Other crops	Translog	2021–22	Cuddalore	Vellore	0.38
Other crops	Translog	2021–22	Cuddalore	Villupuram	0.54
Other crops	Translog	2021–22	Krishnagiri	Thiruvannamalai	0.44
Other crops	Translog	2021–22	Krishnagiri	Vellore	0.47
Other crops	Translog	2021–22	Salem	Thiruvannamalai	0.68
Other crops	Translog	2021–22	Salem	Vellore	0.58
Other crops	Translog	2021–22	Salem	Villupuram	0.53
Other crops	Translog	2021–22	Thiruvannamalai	Villupuram	0.46
Other crops	Translog	2021–22	Vellore	Villupuram	0.47

Table A.13
Number of observations by district for paddy.

District	2017–18	2018–19	2019–20	2020–21	2021–22
Cuddalore	88	96	102	70	75
Dharmapuri	17	3	6	6	3
Krishnagiri	8	8	9	–	–
Salem	3	0	2	18	11
Thiruvannamalai	73	56	101	45	58
Vellore	7	2	4	44	47
Villupuram	11	9	18	30	22

Table A.14
Number of observations by district for sugarcane.

District	2017–18	2018–19	2019–20	2020–21
Cuddalore	27	38	26	29
Dharmapuri	2	3	1	34
Krishnagiri	5	9	10	–
Salem	–	–	–	1
Thiruvannamalai	24	19	13	18
Vellore	2	4	1	7
Villupuram	14	11	5	1

Table A.15
Number of observations by district for other crops.

District	2017–18	2018–19	2019–20	2020–21	2021–22
Cuddalore	49	70	70	82	80
Dharmapuri	23	15	35	5	1
Krishnagiri	12	3	4	29	29
Salem	22	20	20	9	15
Thiruvannamalai	31	44	41	47	28
Vellore	19	25	24	11	18
Villupuram	33	43	53	16	16

Data availability

Data will be made available on request.

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